Technical qualification.

Product information.

Advanced vehicle electrical sys.



BMW Service

General information

Symbols used

The following symbol is used in this document to facilitate better comprehension or to draw attention to very important information:



Contains important safety information and information that needs to be observed strictly in order to guarantee the smooth operation of the system.

Information status and national-market versions

BMW Group vehicles meet the requirements of the highest safety and quality standards. Changes in requirements for environmental protection, customer benefits and design render necessary continuous development of systems and components. Consequently, there may be discrepancies between the contents of this document and the vehicles available in the training course.

This document basically relates to the European version of left hand drive vehicles. Some operating elements or components are arranged differently in right-hand drive vehicles than shown in the graphics in this document. Further differences may arise as the result of the equipment specification in specific markets or countries.

Additional sources of information

Further information on the individual topics can be found in the following:

- Owner's Handbook
- Integrated Service Technical Application.

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The information contained in this document forms an integral part of the BMW Group Technical Qualification and is intended for the trainer and participants in the seminar. Refer to the latest relevant information systems of the BMW Group for any changes/additions to the technical data.

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Advanced vehicle electrical sys.

1. Introduction.

Documents from the Electrical System ICT (**I**nitial **C**ertification **T**raining) give you an overview over the structure and special features of the general vehicle electrical system in BMW and MINI vehicles. With this learning content and the respective qualifications, you can perform general repair work on the general vehicle electrical system under supervision and with the guidance of the workshop information system.



This document provides information designed to give you an overview and contribute towards a better understanding of the qualification content. **As a general principle, the instructions and notes in the workshop information system must be observed for all maintenance and repair work.**

Advanced vehicle electrical sys.

2. Air conditioning.

Humans feel comfortable at a certain ambient temperature and air humidity; they perceive this as their comfort zone. Well-being has a major influence on unimpaired driving capability. The “climate in the vehicle” has a direct influence on the driver, in terms of fatigue-free driving and thus on driving safety.



TE15-1957

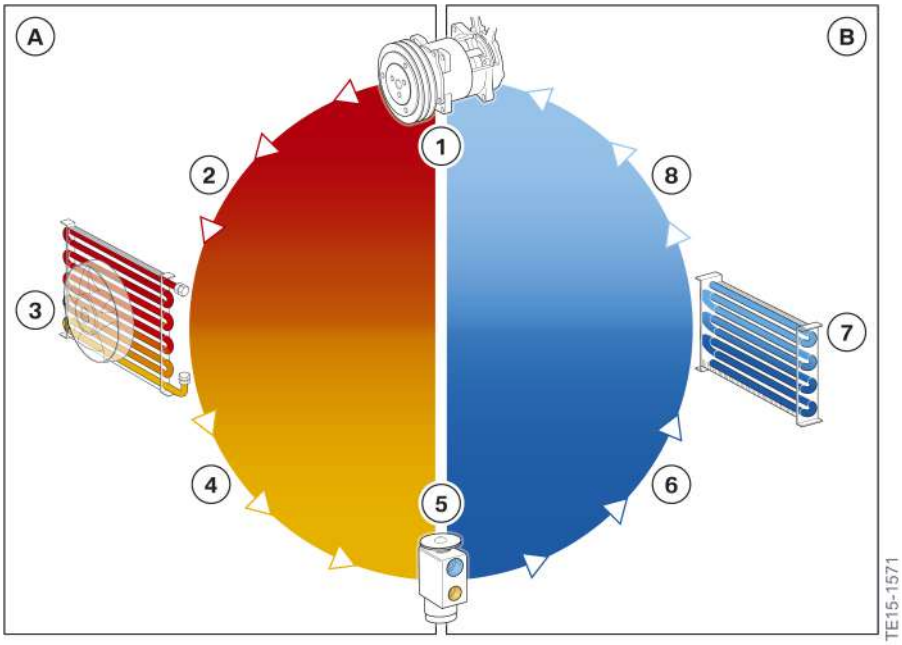
Hygrometer/thermometer

2.1. Principle of the refrigerant circuit

When a gas is compressed, its pressure and temperature increase. If a pressurised refrigerant can expand, it evaporates. The gas needs energy to evaporate, much like a pan with boiling water. The heat required for this is taken from the ambient air. The ambient air, from which the energy is taken, is cooled, and then transported by the blower motor into the vehicle interior as "cold" air. The refrigerant circuit is basically broken down into 4 phases:

- low pressure gaseous
- high pressure gaseous
- high pressure liquid
- low pressure liquid.

2. Air conditioning.



Refrigerant circuit

Index	Explanation
A	High pressure side
B	Low pressure side
1	The compressor increases the pressure and thus also the temperature of the gaseous refrigerant.
2	Gaseous refrigerant at high temperature and high pressure.
3	The air conditioning condenser works like a radiator or heat exchanger. The air flowing past absorbs heat, the hot refrigerant gas cools down and condenses. The refrigerant liquidises.
4	Liquid refrigerant at high temperature and high pressure.
5	The expansion valve expands the refrigerant by means of injection into the evaporator, causing the refrigerant to cool down significantly.
6	Gaseous refrigerant at low temperature and low pressure
7	The evaporator cools and dehumidifies the air flowing past. The refrigerant absorbs heat.
8	Gaseous refrigerant at low temperature and low pressure.

2.2. Description of the refrigerant circuit

An air conditioning system does not produce cold, but conveys heat out of the vehicle interior to the outside.The schematic of the refrigerant circuit shows the functional principle of an air conditioning system. The refrigerant circulates in the closed circuit, continuously transitioning between a liquid and gaseous state while doing so.At the same time, heat is extracted from the vehicle interior and dissipated to the outside.

Advanced vehicle electrical sys.

2. Air conditioning.

The refrigerant circuit is mainly made up of 7 components:

- Compressor
- Air conditioning condenser
- Dryer/intake plenum
- Expansion valve
- Evaporator
- Hoses and pipes
- Open and closed loop control devices.

The components are connected to create a closed refrigerant circuit in which the refrigerant circulates. The air conditioning system works on the principle of a compression refrigeration system, in which the circulating refrigerant is compressed in gaseous state, condenses by releasing heat and evaporates again absorbing heat on reducing the pressure. The refrigerant circuit is divided into a high-pressure part (pressure side) and a low pressure part (intake side).

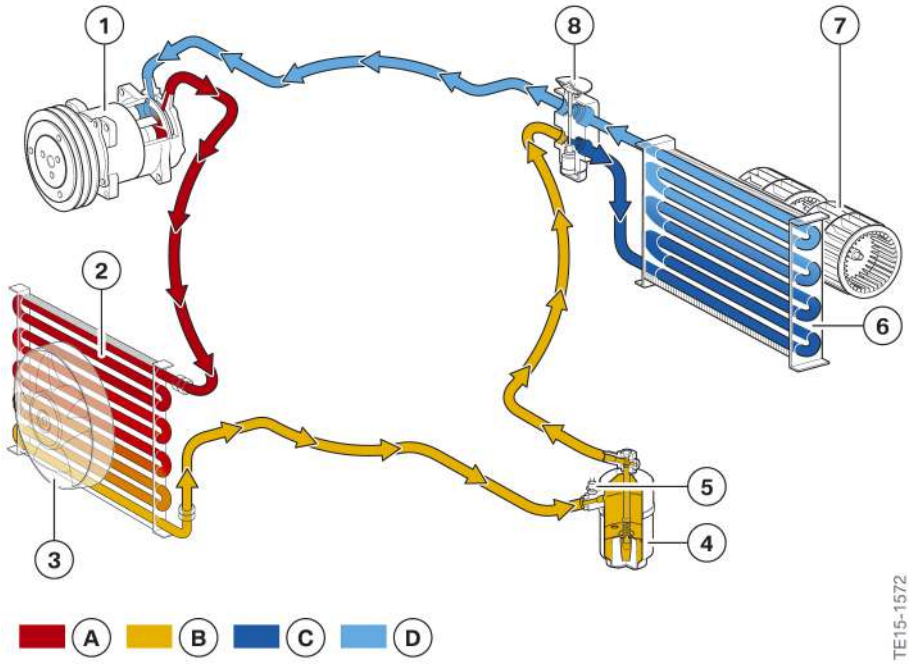
When the refrigerant circuit is activated, that is, when the air conditioning system is switched on with the engine running, the compressor takes in cold, gaseous refrigerant from the evaporator, compresses the refrigerant, which causes it to heat up (max. 120 °C) and pushes it into the air conditioning condenser.The compressed, hot gas is cooled down in the condenser by fresh air flowing past (air stream or rear blower).

On reaching the pressure-dependent dew point, the refrigerant condenses and becomes liquid. The completely liquidised refrigerant, coming from the air conditioning condenser, reaches the fluid tank, where it is collected. In the dryer flask, through which the refrigerant flows, any moisture and contamination is filtered out.

From the fluid tank, the refrigerant flows on to the expansion valve. At this point the high pressure, liquid refrigerant is injected into the evaporator (low pressure part). In the evaporator, the liquid refrigerant expands and evaporates. The evaporation heat required for this is taken from the air flowing past the evaporator fins, thus cooling down the air. The refrigerant, which is now completely gaseous again, is taken in by the compressor and compressed even further. The refrigerant circuit is thus closed.

Advanced vehicle electrical sys.

2. Air conditioning.



Refrigerant circuit with components

Index	Explanation
A	High pressure: gaseous
B	High pressure: liquid
C	Low pressure: liquid
D	Low pressure: gaseous
1	Air conditioning compressor
2	Capacitor
3	Rear blower
4	Fluid tank
5	Pressure sensor
6	Evaporator
7	Evaporator blower
8	Expansion valve

2.2.1. Refrigerant

BMW currently uses 2 different refrigerants.

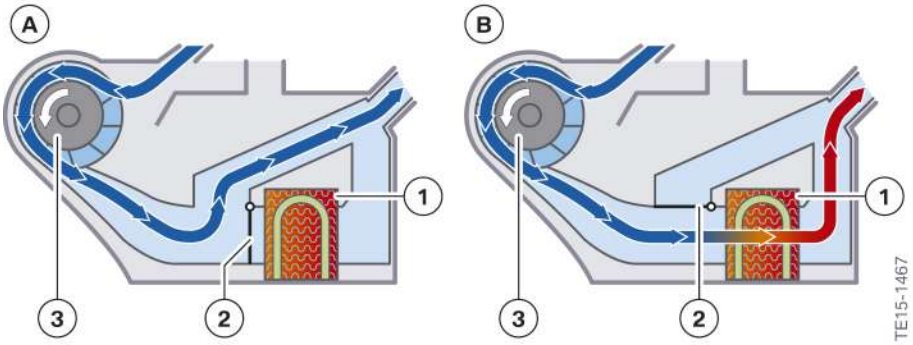
Due to legal requirements of the EU, since January 2011 only refrigerants which have a GWP (Global Warming Potential) below 150 are permitted in all new vehicle types. This shall apply for all newly registered vehicles from 2017. At BMW, the G11/G12 is the first vehicle to be delivered with the new **1234yf** refrigerant. All other vehicles at BMW and MINI use the previous refrigerant, **R134a**.

Advanced vehicle electrical sys.

2. Air conditioning.

2.3. Heater circuit

On current BMW vehicles, the heating is controlled by means of an air-side temperature control. A temperature mixer flap routes the fresh air through a heat exchanger, depending on the selected temperature. The more fresh air flows through the heat exchanger, the hotter it becomes.



Function of the temperature mixer flap

Index	Explanation
A	No heating.
B	Heating.
1	Heat exchanger
2	Temperature mixer flap
3	Blower motor

2.4. Integrated automatic heating/air conditioning system

In addition to manual adjustment of the air conditioning, an integrated automatic heating/air conditioning system available as optional equipment. In an integrated automatic heating/air conditioning system, the temperature and the air distribution as well as the blower intensity are automatically controlled by a control unit.

2.4.1. Automatic air recirculation control sensor

An air quality sensor (automatic air recirculation control sensor) is installed in the air intake area of the engine compartment for automatic air recirculation function. The sensor measures the gases NO_x, CO) of the fresh air drawn in, and automatically switches to air recirculation function on reaching certain limit values.

Advanced vehicle electrical sys.

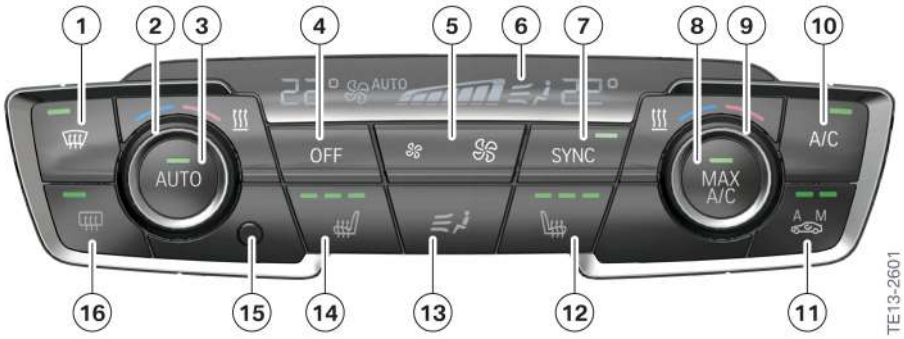
2. Air conditioning.



Automatic air recirculation control sensor installation location

2.4.2. Control panel

In the case of the integrated automatic heating/air conditioning system, the driver sets the desired temperature and blower intensity. The integrated automatic heating/air conditioning system automatically adjusts and manages the air distribution and the blower output.



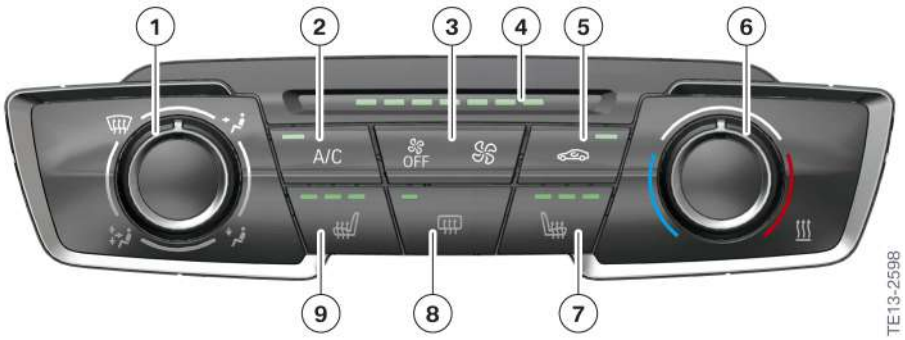
Heating/air-conditioning control panel for integrated automatic heating/air conditioning system

Index	Explanation
1	Defrost window glass and remove condensation
2	Temperature, left
3	Automatic program
4	Switch off system
5	Amount of air, AUTO intensity
6	Display
7	SYNC program; connects the temperature setting on the left and right.

Index	Explanation
8	Maximum cooling
9	Temperature, right
10	Air conditioning function
11	Automatic air recirculation control/air recirculation function
12	Seat heating, right
13	Manual air distribution
14	Seat heating, left
15	Passenger compartment temperature sensor
16	Heated rear window

2.5. Heating/air conditioning control

In the case of the integrated heating and air conditioning control, the driver needs to adjust the heating/air conditioning mechanically. The temperature setting, the blower output and the air distribution are thus adjusted to suit the driver's wishes. There is **no** automatic control.



Heating/air conditioning control panel integrated heating and air conditioning control

Index	Explanation
1	air distribution
2	Air conditioning function
3	Amount of air
4	Display of amount of air
5	Air recirculation function
6	Temperature
7	Seat heating, right
8	Heated rear window
9	Seat heating, left

Advanced vehicle electrical sys.

3. Exterior lights.

BMW vehicles currently have 3 different headlight types. These are:

- Halogen headlights
- Xenon headlights
- LED headlights.

3.1. Halogen headlights

In the case of halogen headlights, low-beam and high beam are implemented by means of bulbs. Bulbs or LEDs are installed to provide the side light or daytime running light functions, depending on the vehicle. In the following graphic, you can see a halogen headlight, based on the example of a F20:



Halogen headlightsF20

Index	Explanation
1	Turn indicator
2	Daytime driving lights
3	Side lights
4	High-beam headlight
5	Low-beam headlight

3.2. Xenon headlight

In the case of xenon headlights, a distinction is made between:

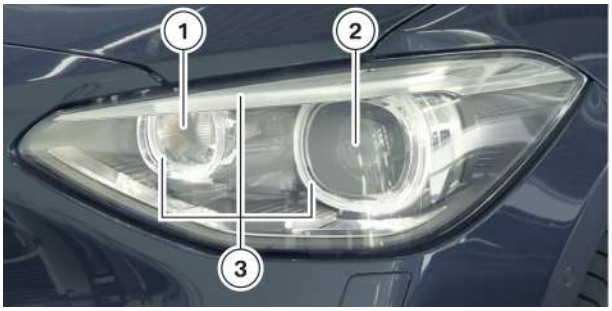
- Xenon headlights
- Bi-xenon headlights

In the case of bi-xenon headlights, a shared xenon bulb is used for low and high beam. A finisher built into the headlight is used to change over between dipped and high beam. Bi-xenon headlights have been exclusively used since E46.

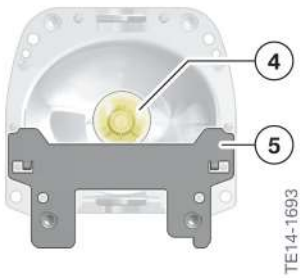
The following graphic shows a bi-xenon headlight from a F20, along with its components:

Advanced vehicle electrical sys.

3. Exterior lights.







Bi-xenon headlightF20

Index	Explanation
1	Turn indicator
2	Low-beam and high-beam headlight
3	Side and daytime riding light
4	Xenon bulb
5	Finisher for changing over between low and high beam

3.3. LED headlights

LED headlights have been used by BMW since the F12/F13. The G11/G12 already has LED headlights as part of its standard equipment.

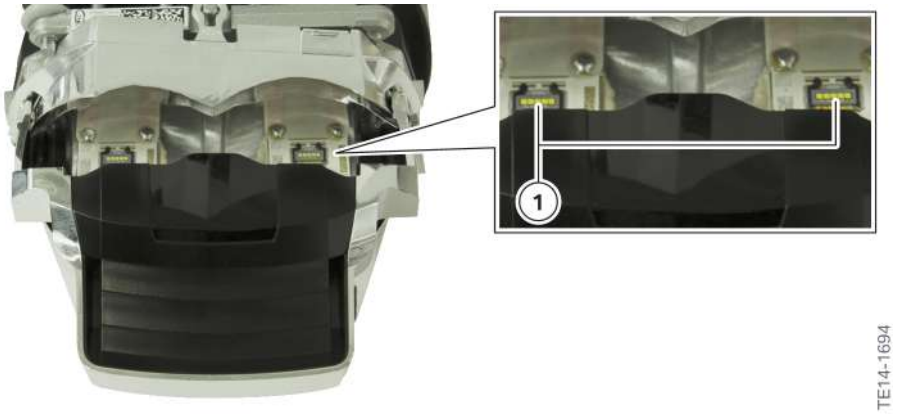


F01 LCI adaptive LED headlights

Because the individual LEDs generate a lower light quantity compared with bulbs or xenon bulbs, multiple LEDs are typically bundled and interconnected to form LED modules. In contrast to xenon headlights, LED light has an even "whiter" light that is becoming increasingly similar to daylight.

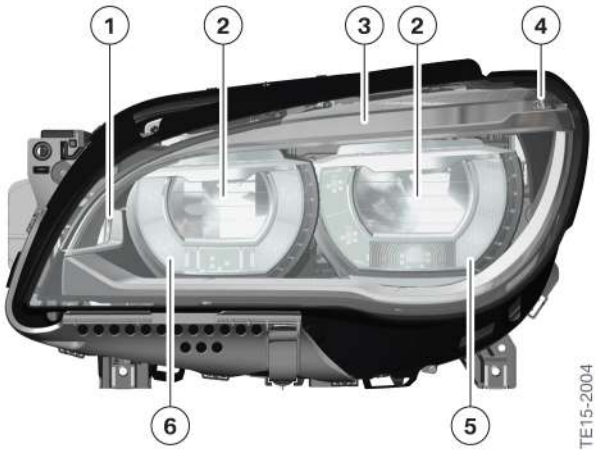
The benefits of LED headlights include:

- Long service life
- High efficiency
- Mechanical robustness
- Adaptation of the light colour
- Possibility of dimming.



LED module

3.3.1. Design

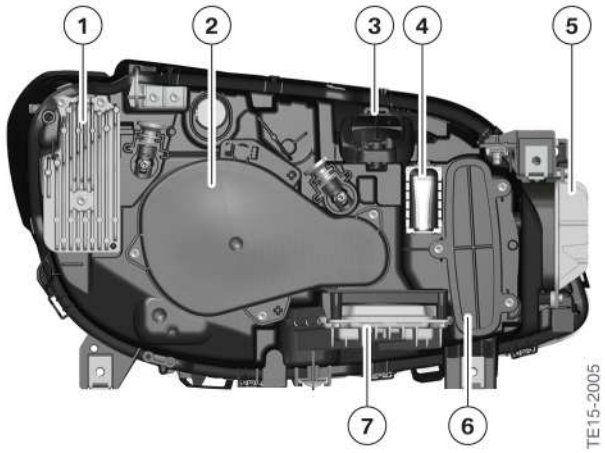


F01 LCI components LED headlight front view

Index	Explanation
1	Cornering lights
2	High-beam headlight
3	Turn indicator and positioning light/side lights
4	Side marker light
5	Low-beam headlight/Daytime driving lights
6	Low-beam headlight/Daytime driving lights

Advanced vehicle electrical sys.

3. Exterior lights.



F01 LCI components LED headlight rear view

Index	Explanation
1	LED module for turn indicator and positioning light/side lights
2	Rear cover (outer fan, stepper motors)
3	Cover at top (inner fan)
4	Electrical plug connection
5	Cornering light LED module
6	Headlight driver module STML/STMR
7	LED main light module LHML/LHMR

LED headlights require external fans to cool the LED modules .

3.3.2. Headlight cleaning system

A headlight cleaning system is a spraying device for cleaning the headlights on vehicles. The washer fluid is taken from the wash/wipe system collection fluid tank via a separate pump.

A headlight cleaning system is mandatory for motor vehicles with a luminous flux of more than 2000 lumens. (Lumen is the unit in which the luminous flux is stated (a typical 60-watt domestic bulb has about 400 lumen)).

In vehicles since the F45, or at MINI since the F56, that have LED headlights, a headlight cleaning system is not included as standard equipment.

LED technology helps to comply with the 2000 lumen limit. This limit was not adhered to with bi-xenon headlights. For this reason, a headlight cleaning system is legally required for this lighting technology.

Advanced vehicle electrical sys.

3. Exterior lights.



F30 Headlight cleaning system

3.4. Adaptive headlights

3.4.1. Adaptive cornering light

Exterior lighting with adaptive headlights is available as optional equipment for current BMW models. Adaptive headlights are only available with bi-xenon headlights or LED headlights. During cornering, the headlight on the inside of the corner swivels to illuminate the corner in a superior way.

3.4.2. Variable light distribution

Variable light distribution improves the illumination of the road in front of the vehicle. This means that, e.g. objects on the road can be detected earlier.

The following variable light distributions are possible:

- Adaptive headlight including cornering light
- Urban beam pattern
- Single-carriageway beam pattern
- Fog beam pattern
- Motorway beam pattern.

The various light distributions are integrated into the existing lighting functions.

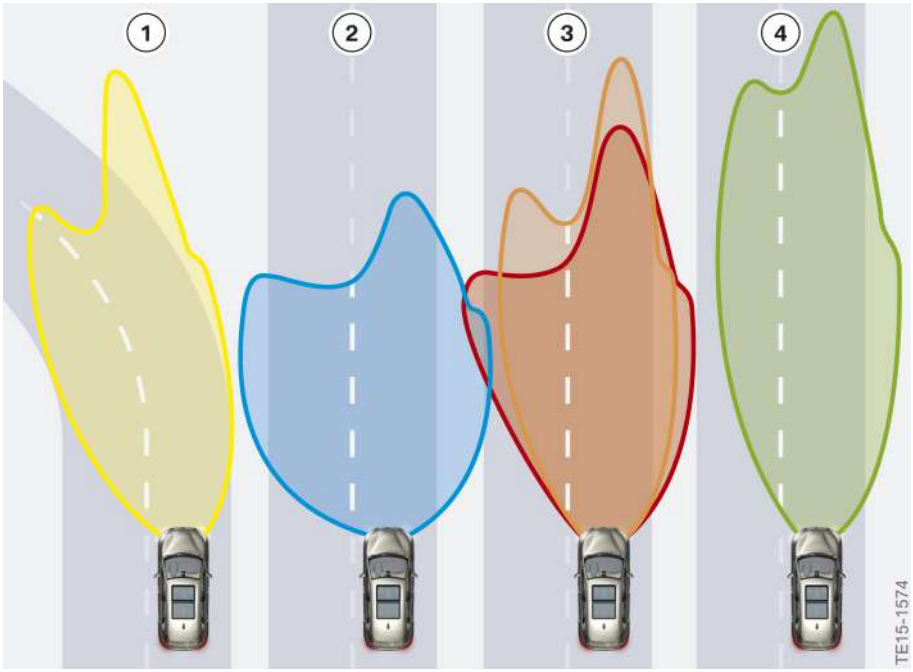
The benefits of variable light distribution are:

- Improved illumination of the road in front of the vehicle, thus reducing the accident risk potential due to identifying traffic situations too late, e.g. earlier detection of objects on the road
- More relaxed and safer driving due to better vision in front of the vehicle
- Improved illumination in urban driving

Advanced vehicle electrical sys.

3. Exterior lights.

- Increased headlight beam throw for motorway driving
- Illumination of the inside of the bend up to approx. 70 km/h
- Improved illumination in front of the vehicle after switching on the fog lights due to the wider lighting spread.



Road illumination

Index	Explanation
1	Adaptive cornering light
2	Fog beam pattern
3	Single-carriageway beam pattern (orange)/urban beam pattern (red)
4	Motorway beam pattern

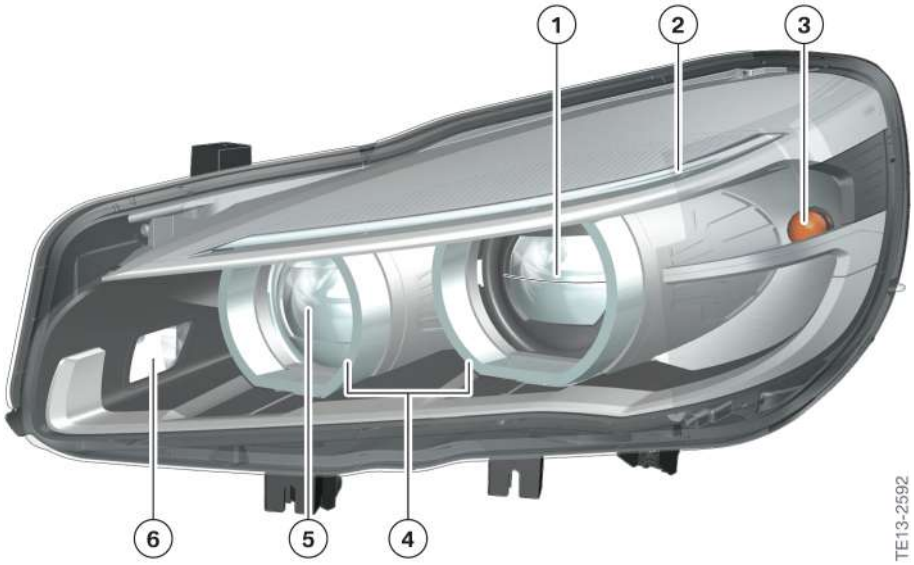
The single-carriageway beam pattern is equivalent to the "normal" low-beam headlight setting.

Static adaptive headlight

MINI vehicles (as of F56), the F45, F46 and the F48 have static cornering light in combination with the optional equipment “LED headlight with extended packages". The LEDs responsible for cornering light are actuated in addition to the low-beam headlight during cornering, in order to illuminate the bend.

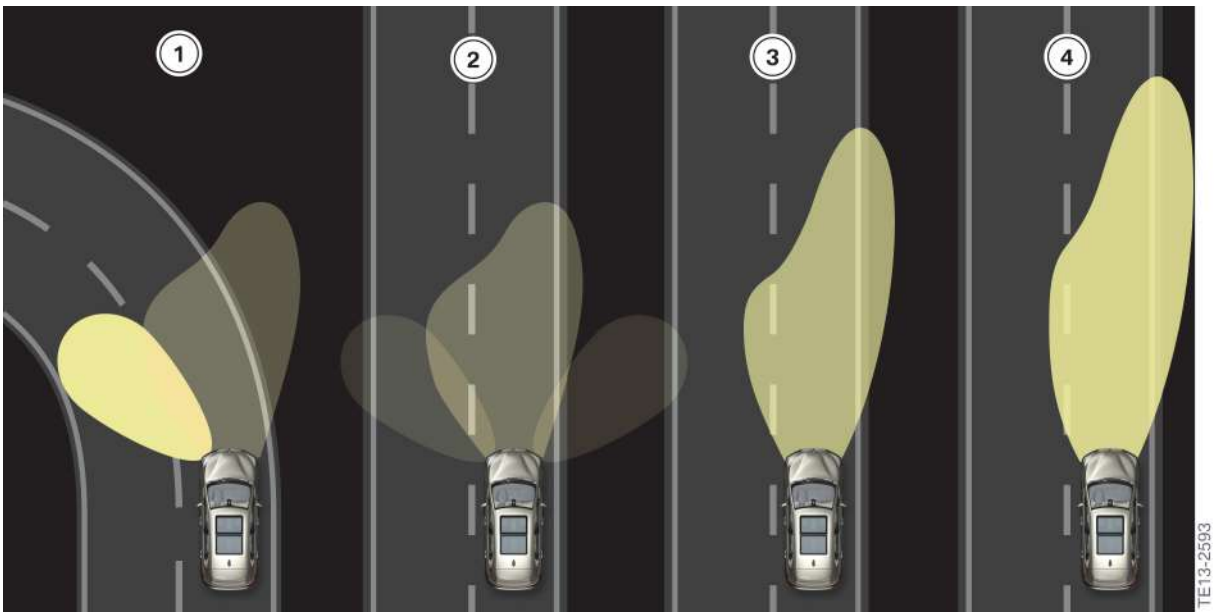
Advanced vehicle electrical sys.

3. Exterior lights.



F45 LED headlight with extended scope

Index	Explanation
1	Low-beam headlight/High-beam headlight
2	Accent light
3	Bulb for turn indicator
4	Side lights/Daytime driving lights
5	Design trim
6	Cornering lights



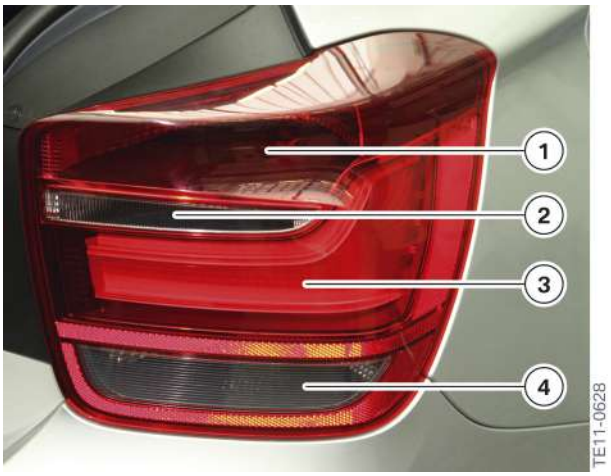
F45 Adaptive light distribution

Index	Explanation
1	Cornering lights
2	Urban beam pattern
3	Single-carriageway beam pattern
4	Motorway beam pattern

3.5. Rear light

Depending on the vehicle, single- or two-row rear lights are installed. These lights can be further distinguished depending on the equipment level.There are rear lights that exclusively have bulbs installed, and rear lights where the light sources are partially or completely LEDs.

In the following graphic, you can see a rear light, based on the example of F20:



F20 rear light

Index	Explanation
1	Brake light
2	Turn indicator
3	Tail light
4	Rear fog light/reversing light

3.6. Light setting

The exterior lights is an important element of active safety. In addition to the legal requirements, modern BMW vehicles also have numerous functions depending on the equipment and national-market version, in order to optimise the perception, and thus the safety and comfort, of the driver.

In order to guarantee the proper function of the headlights, the correct procedure for the adjusting procedures must be observed. This is described in the respective repair instructions.

Advanced vehicle electrical sys.

3. Exterior lights.

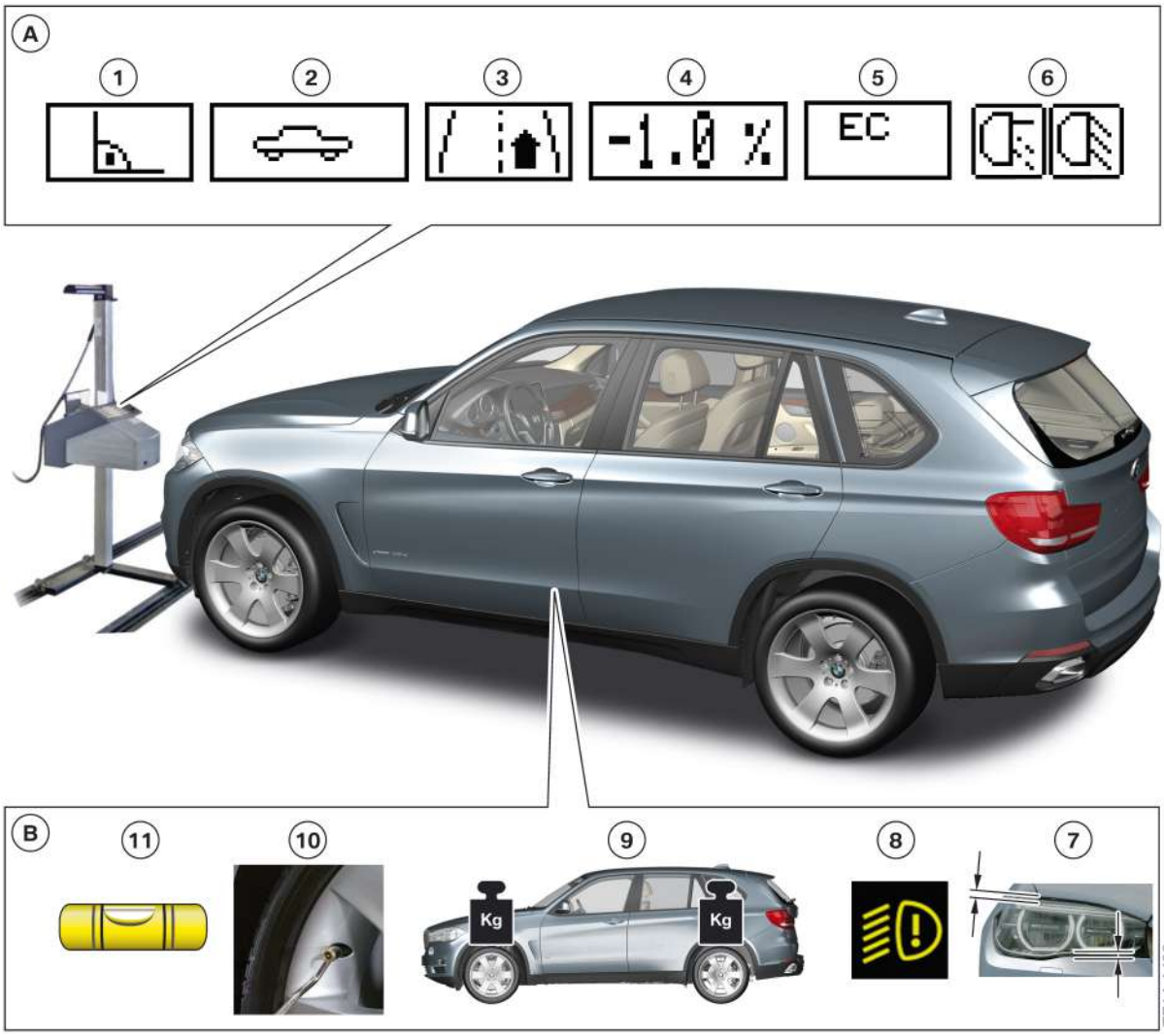
The light setting is effected using adjusting screws. Using a headlight adjusting device, the actual and setpoint values can be compared for the respective lighting function.

The headlights must be adjusted after the following work:

- Replacement of the headlight
- Slackening of the mounting bolts
- Modification of the vehicle height (for e.g. low-slung).

3.6.1. Requirement

Before the light setting, some general prerequisites must be satisfied at the vehicle and at the headlight adjusting device.



Prerequisites for the light setting

Index	Explanation
A	**Prerequisites at the headlight adjusting device**
1	Correct alignment of headlight adjusting device to vehicle
2	Correct vehicle type
3	Correct selection of right-hand traffic/left-hand traffic
4	Correct setting dimension for light/dark boundary
5	Correct headlight type (halogen, xenon, LED)
6	Correct type of lighting effect of headlight (only for low-beam headlight and high beam)
B	**Prerequisites at the vehicle**
7	Correct installation dimensions of the headlights
8	All light sources OK
9	Vehicle load as per the repair instructions
10	Correct tyre pressure
11	Vehicle is on an even surface

Depending on the equipment of the vehicle and version of the headlight adjusting device, other prerequisites may also have to be satisfied.

3.6.2. Headlight beam throw adjustment

If the vehicle has a manual headlight beam throw adjustment, the handwheel must be set to zero before the adjustment.

If the vehicle has an automatic headlight beam throw adjustment, a waiting period of 80 seconds must be observed after the light has been switched on. During this time the vehicle cannot be moved.

3.6.3. Light switch

In order to adjust the low-beam headlight, the light switch must be in the "low-beam headlight/driving light" position and **not** in the "automatic driving lights control" position.



Light operating unit

Advanced vehicle electrical sys.

3. Exterior lights.

Index	Explanation
1	Automatic driving lights control
2	Side lights
3	Low-beam and driving light



For the adjustment of the headlight mechanics, it is absolutely imperative that the procedure described in the repair instructions and the information in the operating instructions of the headlight adjusting device are observed.

The headlight mechanics adjustment cannot be effected immediately after lowering the vehicle on the vehicle hoist. In this case, the vehicle must be moved approx. 15 metres on its own axles before adjusting the light setting.

Advanced vehicle electrical sys.

4. Vehicle access.

4.1. Central locking system

All BMW and MINI vehicles are equipped with a central locking system as standard. Besides the doors, the system also controls the tailgate and fuel filler flap.

The central locking system is controlled via the following components:

- ID transmitter
- Lock barrel
- Central locking button
- Outside door handle (only with optional equipment “Comfort Access”).

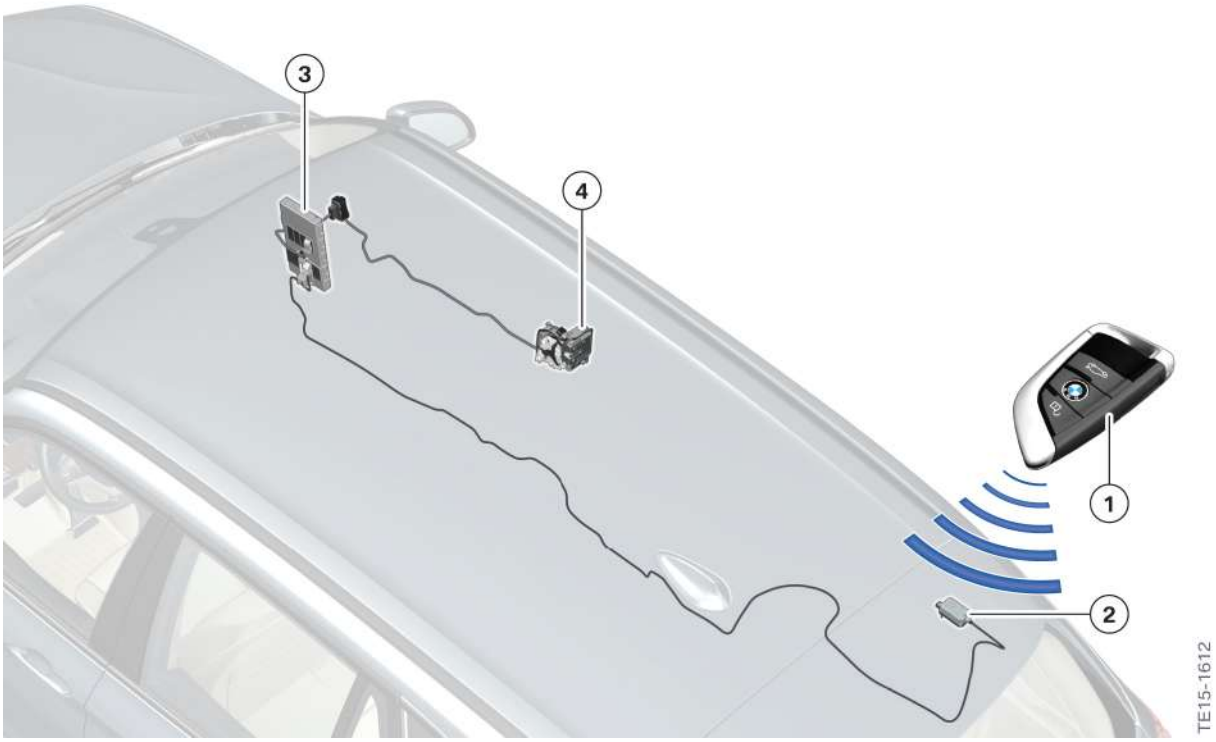


ID transmitter

Index	Explanation
1	Unlocking/convenient opening
2	Locking/convenient closing
3	Tailgate
4	Freely assignable button

4.1.1. Unlocking

When release button on the ID transmitter is pressed, the transmitter outputs an encrypted signal. This signal is received by the remote control receiver, converted and forwarded to the control unit responsible for the central locking system. After the control unit has identified the ID transmitter that matches the vehicle, the vehicle is unlocked.



Central locking system function

Index	Explanation
1	ID transmitter
2	Remote control receiver
3	Control unit for central locking system; here in the F15 a Body Domain Controller, BDC
4	Door lock

4.1.2. Lock

The vehicle can only be locked after evaluating the door contacts and if the driver's door is closed. Locking follows the same steps as unlocking. The ID transmitter sends an encrypted signal to the remote control receiver. The receiver receives the signal, converts it, and forwards it to the appropriate control unit.

4.1.3. Central locking button

The vehicle can be locked or unlocked with the central locking button.

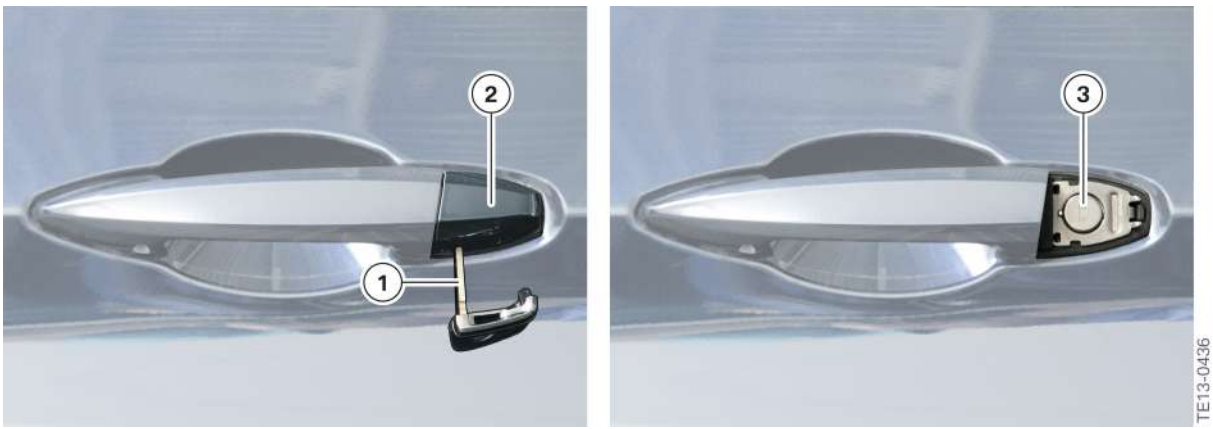


Button for central locking system

Index	Explanation
1	Button for central locking system (unlocking)
2	Button for central locking system (locking)
3	Tailgate pushbutton

4.1.4. Lock barrel

In case of a faulty central locking system or ID transmitter, there is the option of locking or unlocking the vehicle using the mechanical key.



Outer door handle

Index	Explanation
1	Mechanical key
2	Cover
3	Lock barrel

Advanced vehicle electrical sys.

4. Vehicle access.

4.2. Comfort Access

The optional equipment Comfort Access enables access, starting and locking of the vehicle without actively operating the ID transmitter. You only need to have the ID transmitter with you.

For example, pulling the outside door handle triggers a pulse via the sensor. The outside door handle electronics evaluate the sensor and report that vehicle access has been requested. For MINI, a button on the outside door handle must be pressed to unlock or lock the vehicle.

Comfort Access contains the following functions:

- **Unlocking**
  This function supports keyless vehicle access if a valid ID transmitter is in the immediate vicinity of the vehicle.
- **Engine start**
  The function enables starting of the vehicle if there is a valid ID transmitter in the passenger compartment.
- **Locking**
  This function supports keyless locking of the vehicle if a valid ID transmitter is in the immediate vicinity of the vehicle.

4.2.1. Ring aerial

Vehicles as of the F series have the "keyless start" function as standard equipment. In case of a technical fault, or a flat battery in the ID transmitter, the transmitter is no longer identified in the vehicle interior, and the vehicle cannot be started. For this reason, a ring aerial is installed in the vehicle' steering column. To start the engine, the ID transmitter is held at the marked position on the steering column while pressing the start/stop button at the same time.



F45 Ring aerial

Index	Explanation
1	Ring aerial

For vehicles of the E series with optional equipment “Comfort Access”, the ID transmitter must be inserted into the insertion slot in case of a technical fault, or a flat battery in the ID transmitter.

Advanced vehicle electrical sys.

5. Seats.

BMW vehicles have a variety of seat versions and seat adjustment options.

The following seat versions are currently available:

- Basic seat
- Sports seat
- Comfort seat.

Additionally, a distinction is made as to whether the seats are mechanically or electrically adjustable.

In the case of mechanically adjustable seats, seat adjustment is performed mechanically using lever adjusters. In the case of electrically adjustable seats, the seat is adjusted by electric motors. There are some seats where part of the adjustment must be performed mechanically, and the other part electrically (forward/back seat adjustment mechanical, backrest angle adjustment electrical).

5.1. Electrically adjustable seats

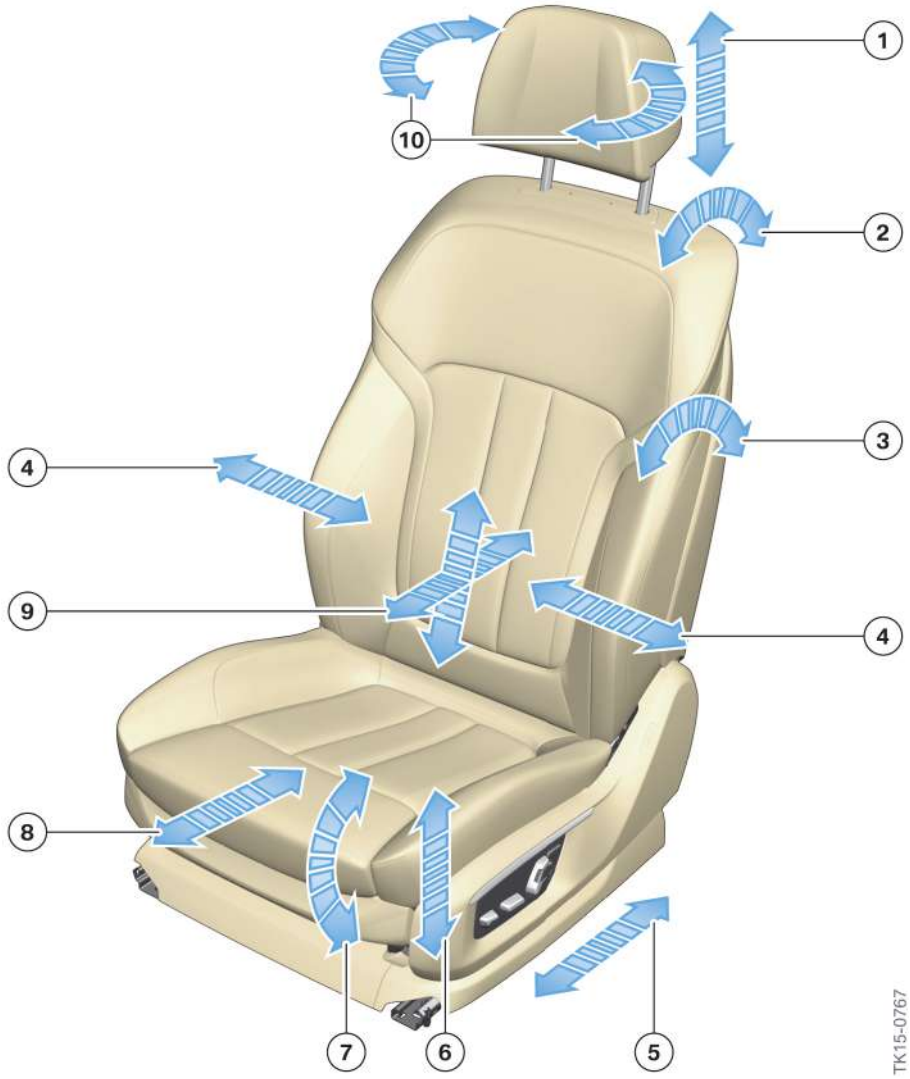
Electrically adjustable seats are optional equipment in most vehicles. Depending on the vehicle and equipment, either only the front seats are electrically adjustable, or also the rear seats.

5.1.1. Seat adjustment options

The adjustment options are equipment- and vehicle-dependent, however. The following graphic shows you the seat adjustment options for a maximum equipment level

Advanced vehicle electrical sys.

5. Seats.

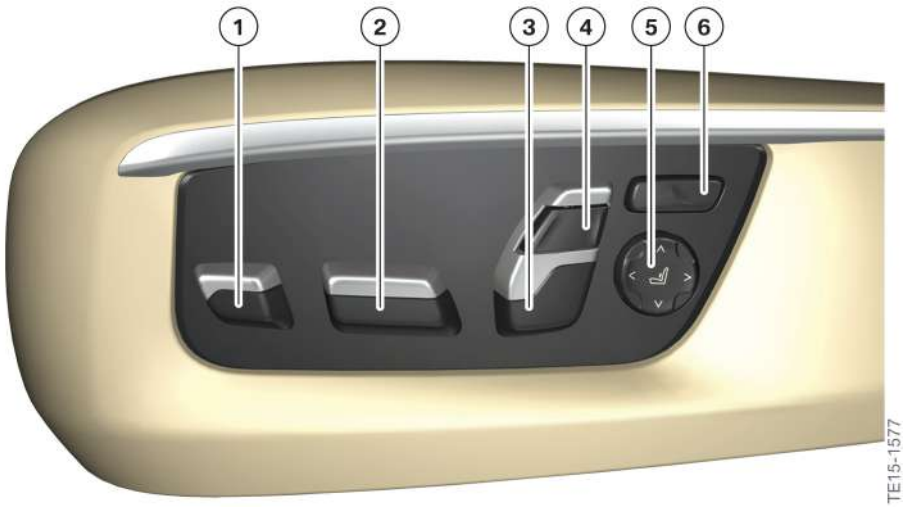


G11/G12 Maximum seat adjustment options

Index	Explanation
1	Head restraint height adjustment
2	Backrest upper section adjustment
3	Backrest angle adjustment
4	Backrest width adjustment
5	Forward/back seat adjustment
6	Seat height adjustment
7	Seat angle adjustment
8	Seat depth adjustment
9	Lumbar support
10	Head restraint side bolster adjustment (mechanical)

5.1.2. Seat adjustment switch

The seats are adjusted via the switch for fore-and-aft seat adjustment. The switch for fore-and-aft seat adjustment is installed laterally on the seat



Seat adjustment switch

Index	Explanation
1	Seat depth adjustment
2	Forward/back, seat height and seat angle adjustment
3	Backrest angle and head restraint height adjustment
4	Backrest upper section adjustment
5	Lumbar support adjustment
6	Backrest width adjustment

5.1.3. Memory function

The memory button allows two driver's seat and mirror positions, and, depending on the equipment level, the steering wheel position, to be saved and loaded.

There are three options for calling up the settings.

Briefly press the button with the desired seat adjustment upon exiting the vehicle. For this purpose, the driver's door must be open and terminal 15 must be off.

The driver's door must be closed and terminal 15 must be on when getting into the vehicle. Press the button for the desired seat adjustment until the process has completed.

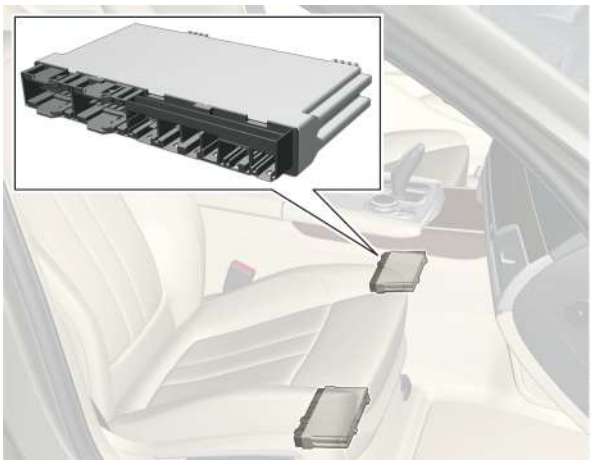
The seat, and possibly the steering wheel, are moved to the stored position via a stored profile on unlocking the vehicle.

Advanced vehicle electrical sys.

5. Seats.

5.1.4. Control unit

The control units for electrical seat adjustment are located underneath the respective seats. The control units control the optional equipment, electrical seat adjustment, memory function, seat heating, active seat ventilation and the massage functions.



Control unit electrical seat adjustment driver's side and passenger's side

Advanced vehicle electrical sys.

6. Alarm system.

An alarm system can be ordered as optional equipment. The alarm system's task is to alert in case of unauthorised access to the vehicle. The alarm system monitors the complete vehicle interior. Beyond this, the engine compartment and vehicle stationary state are monitored.

6.1. Control unit

The ultrasound interior movement detector of the alarm system is located in the roof function centre.The emergency power siren with tilt sensor is located in the rear wheel arch.



G11/G12 roof function centre

Index	Explanation
1	Roof function centre



Siren with tilt alarm sensor

Advanced vehicle electrical sys.

7. Assist systems.

Numerous assistance systems support the driver in various ways, thus increasing the comfort and safety of all occupants. The different technologies are assigned to an exact detection of the surrounding area.

This product information gives you an overview of the assistance systems and their sensors in current BMW vehicles.

The following table gives you an overview of the assistance systems available at BMW, including their sensors, and a brief description (KAFAS = camera-based driver assistance systems):

System	Sensor	Description
Cruise control with braking function	Wheel speed sensor 	The cruise control with braking function keeps the vehicle at the speed set by the driver.
Speed limit	Wheel speed sensor 	A speed between 30 and 230 km/h can be set using the speed limit function. The set speed can only be overridden by an escape function by fully depressing the accelerator pedal.
High-beam assistant (FLA)	KAFAS camera or high-beam assistant camera 	The high-beam assistant detects other vehicles at an early stage and switches to high- or low-beam depending on the traffic situation. In contrast to this, the optional equipment “BMW Selective Beam” selectively masks other road users out of high-beam headlight distribution at a headlight beam throw of 400 m.

Advanced vehicle electrical sys.

7. Assist systems.

System	Sensor	Description
Road sign recognition	KAFAS camera 	Road sign detection including no-overtaking indicators is an assistance system that detect current and variable top speed limitations and information on overtaking prohibitions and displays them to the driver in the instrument cluster or in the optional Head-Up Display.
Collision warning with city braking function	KAFAS camera 	At speeds up to 60 km/h, the driver is not only warned, but braking is initiated to avoid a collision.
Pedestrian warning with city braking function	KAFAS camera 	Pedestrians are also detected, and braking is initiated to avoid a collision at speeds up to 60 km/h.
Lane departure warning	KAFAS camera 	Lane departure warning registers lane markers over a distance of 50 m and as of a speed of around 70 km/h. On driving over a marker, the driver is warned by a light vibration on the steering wheel.
Lane change warning SWW	Radar sensors 	Lane change warning alerts if vehicles in the dead spot, or in the adjacent lane, are approaching at high speed.The system alerts via vibration in the steering wheel and a flashing warning symbol on the exterior mirror.

Advanced vehicle electrical sys.

7. Assist systems.

System	Sensor	Description
Active Cruise Control (ACC) with Stop & Go	Radar sensor and/or KAFAS camera  	Active cruise control with Stop&Go function autonomously keeps the selected distance to the vehicle driving ahead, automatically controls the speed down to a standstill, and automatically accelerates to the desired speed as soon as the traffic flow allows this to happen.
Traffic Jam Assist	Radar sensor and/or KAFAS camera  	The traffic jam assistant relieves the burden on the driver in monotonous traffic situations. The system automatically regulates the distance in dense traffic conditions at speeds of up to around 60 km/h, it automatically adjusts the speed to a standstill and actively steers while doing so.
Tailgate warning	Radar sensor and/or KAFAS camera  	Collision warning detects vehicles ahead and warns in case of an imminent risk of collision.

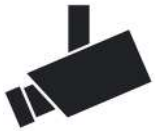
Advanced vehicle electrical sys.

7. Assist systems.

System	Sensor	Description
Night vision	Thermal imaging camera 	BMW Night Vision with Dynamic Light Spot detects people and animals at a far distance, and illuminates them in a targeted way.A thermal imaging camera in the front grille with a range of up to 300 metres relays the images of people and animals directly to the central information display.As soon as Night Vision identifies pedestrians or larger animals on a collision course in the dark, the system picks them out with two separately controllable spotlights without dazzling them in an annoying way.
Park Distance Control (PDC)	Ultrasonic sensors 	Park Distance Control at the front and rear facilitates parking and manoeuvring in a tight space. When the vehicle approaches an obstacle, warning signals are output and display on the central information display.

Advanced vehicle electrical sys.

7. Assist systems.

System	Sensor	Description
Parking manoeuvring assistant PMA	Ultrasonic sensors 	The Parking Manoeuvring Assistant facilitates park in parking spaces. The system measures suitable parking spaces while driving past.After finding a suitable parking space, the Parking Manoeuvring Assistant takes over the steering, and possibly also accelerating and braking.
Surround view	Cameras 	A wide angle front camera, two cameras below the exterior mirrors, and a wide angle rear view camera provide a 360° view of the vehicle's environment for easier manoeuvring and parking.
Rear view camera (RFK)	Camera 	The rear view camera provides better orientation on reversing in parking situations and in the speed range below 15 km/h. It shows the area behind the vehicle on the central information display. Interactive markers for distance and turning circle support the driver during manoeuvring. Obstacles are displayed with coloured marking.

Advanced vehicle electrical sys.

7. Assist systems.

7.1. Sensors and cameras



F15 sensors, outer

Index	Explanation
1	Front radar sensor
2	KAFAS camera
3	Reversing camera
4	PDC ultrasonic sensor (Park Distance Control)
5	Rear radar sensor (right and left)
6	Top view camera (right and left)
7	PMA ultrasonic sensor (Parking Manoeuvring Assistant)
8	Night Vision camera
9	Front camera

The number, version and installation position of the sensors and cameras may vary significantly depending on the model and equipment of the respective vehicle. The technology of the sensors and cameras is also constantly undergoing changes due to the further development of the assistance systems.

7.1.1. Subsequent work

Following repairs, additional work on the vehicle electronics is often required. Different measures may be required, depending on whether a sensor or camera has been replaced or only removed. Sometimes these measures must be carried out with help of the workshop information system.

- **Calibration**
  Mechanical or electronic setting of the sensor or camera alignment
- **Initialisation (teach-in)**
  Teach-in of start and end values
- **Encoding**
  Adapting to national-market version, vehicle equipment and vehicle type
- **Programming**
  Renewing the data and program status.

If the necessary work has not been carried out, this may lead to malfunctions or a total failure of the assistance systems. It often also causes fault messages and, as a result, fault memory entries. These entries then need to be processed according to the specifications in the workshop information system.



If the necessary calibrations, initialisations, encoding or programming are not carried out, this may lead to malfunctions or a total failure of the assistance systems.

Advanced vehicle electrical sys.

8. Displays and Controls.

8.1. Head-Up Display

"Head-Up" describes the principle benefit of this system. The Head-Up Display HUD projects a virtual image into the driver's field of view. Important information, such as the cruise control or notes from the arrow display are reflected on the windscreen, and thus permanently visible in the vehicle environment.

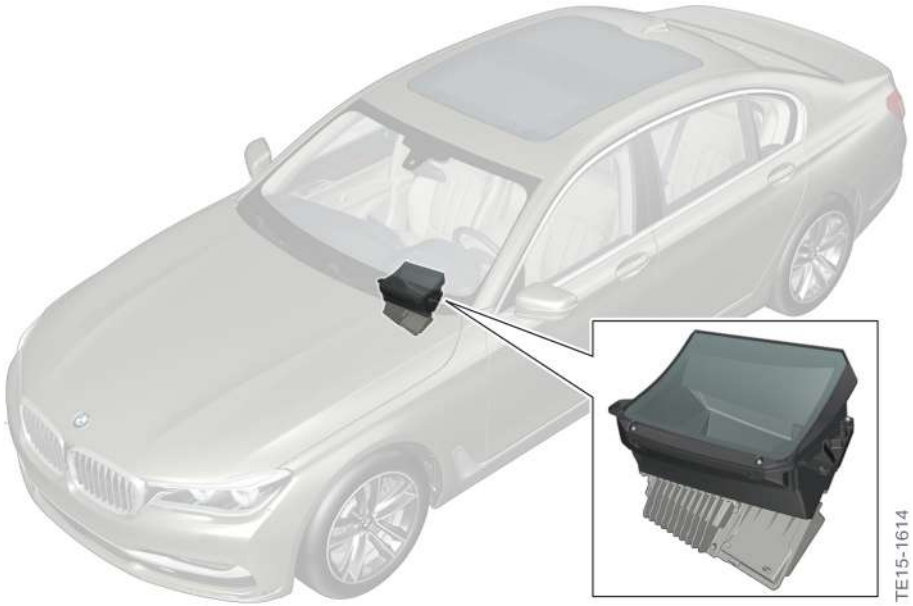
The Head-Up Display contains various functions aimed at enhancing road safety and driving comfort. This includes the following displays:

- Cruise control
- Navigation system
- Check Control messages
- Speed
- Multimedia.



Head-Up Display (HUD)

The installation location of head-up display is above the steering column, immediately behind the instrument cluster.



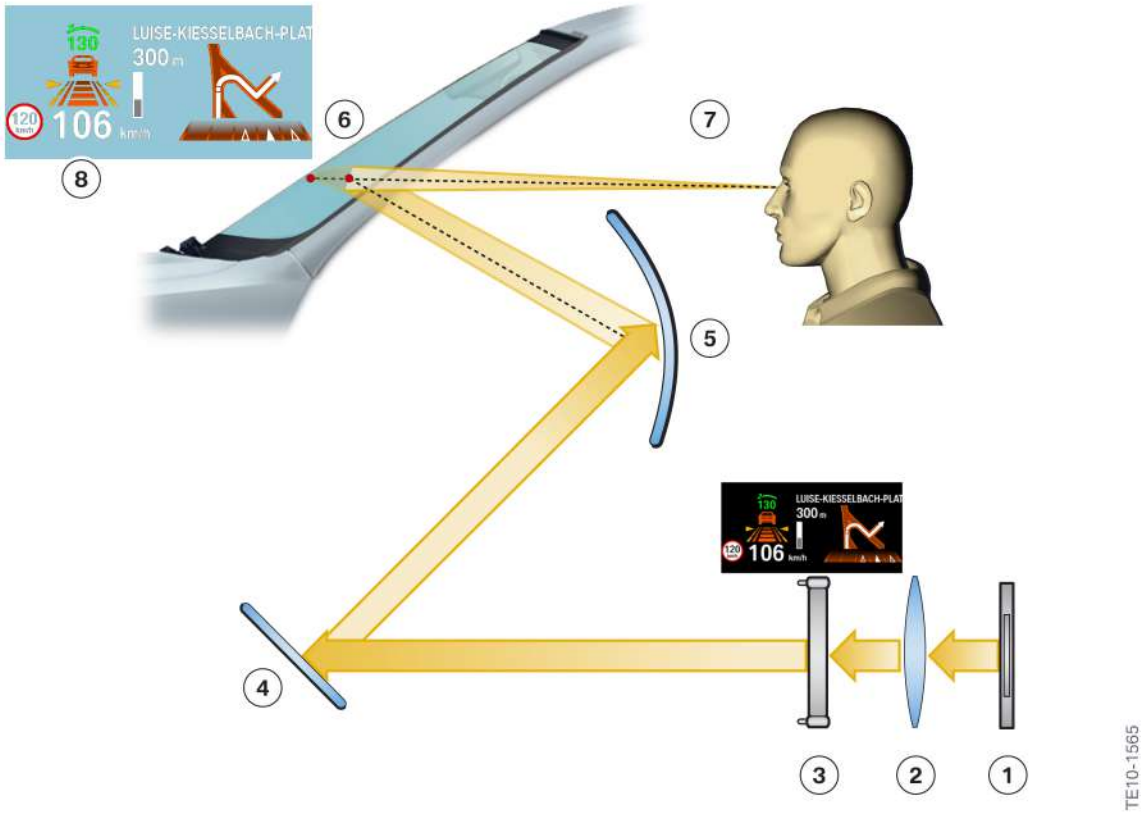
G11/G12 Head-Up Display HUD installation location

8.1.1. Function

The HUD is similar to a projector. The light source is provided by LEDs. The screen content is formed via a TFT projection display. The windscreen acts like a reflecting mirror. Due to this type of projection, the image looks as though it is "floating" in front of the vehicle from the driver's point of view.

Advanced vehicle electrical sys.

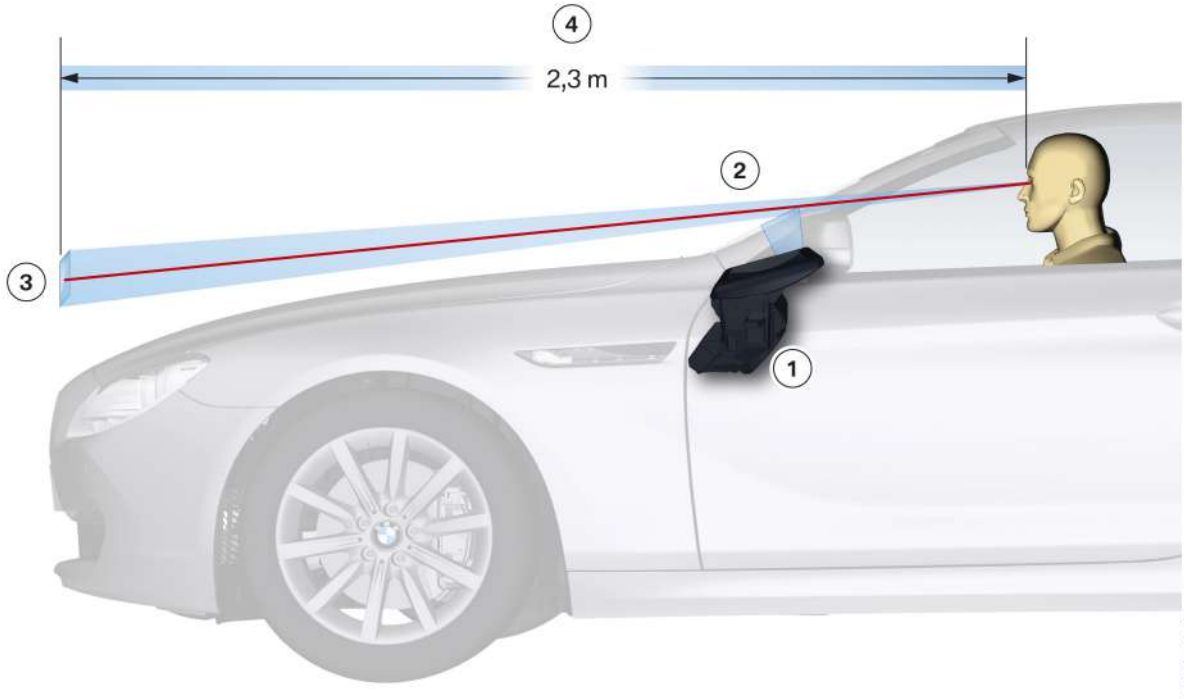
8. Displays and Controls.



Function of Head-Up Display

Index	Explanation
1	Light source
2	Lens
3	TFT projection display
4	Plane mirror
5	Curved mirror
6	Windscreen
7	Observer's point of vision
8	Projected image

The projected HUD image content appears to be approximately 2.3 m from the observer's eye.

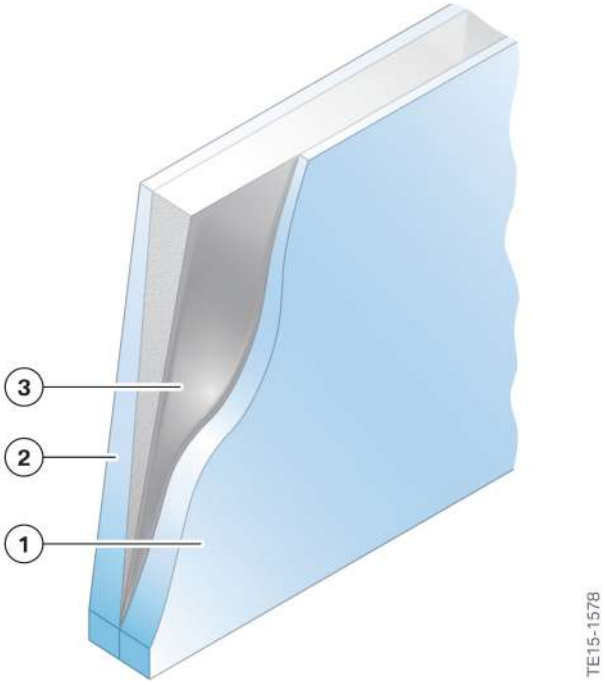


Projection distance

Index	Explanation
1	Head-Up Display HUD
2	Windscreen
3	Projected image
4	Projection distance

Windscreen

The windscreen on vehicles with a Head-Up Display is an important element for projecting the displays. The outer and inner glass panes are bonded to a plastic film, just like in a standard windscreen. In contrast to this, the film is not plane-parallel, but runs across the entire windscreen in a wedge shape.

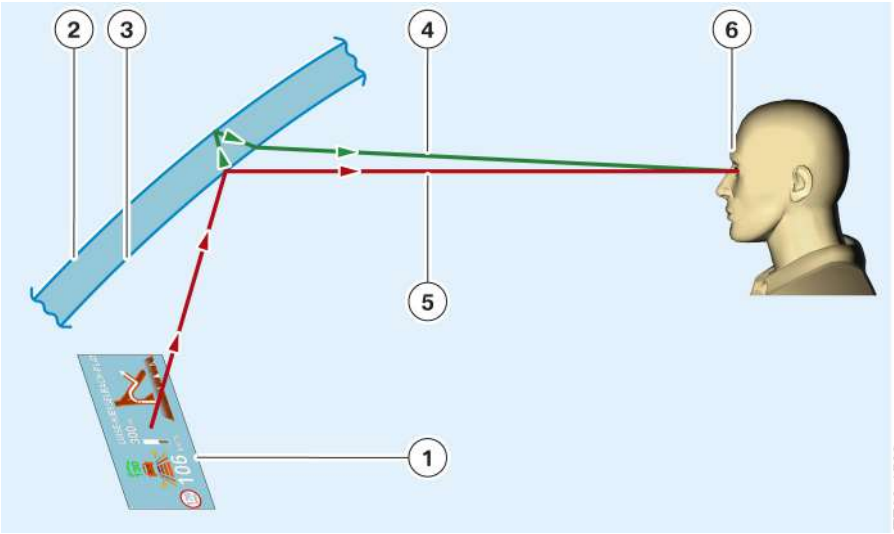


Windscreen HUD

Index	Explanation
1	Inner windscreen
2	Outer windscreen
3	Special foil

The HUD image is always reflected on the inner surface and outer surface of the windscreen. With a standard windscreen, the two reflected images are offset due to the tilt angle of the window glass.

Due to the taper angle, these two images are superimposed on the HUD window glass with the result that the driver only sees "one" image.



Double reflection

Index	Explanation
1	Display
2	Outer surface of the windscreen
3	Inner surface of the windscreen
4	Reflection on the outer surface of the windscreen
5	Reflection on the inner surface of the windscreen
6	Driver's eye

The following graphic shows the extent to which the image is distorted by the windscreen:



Appearance of HUD with standard windscreen

Index	Explanation
1	Image with the right windscreen
2	Image with the wrong windscreen

8.1.2. Combination disc

The F45/F46 and the MINI models (as of F56) have a Head-Up Display with indicator pane due to the installation position of the windscreen. In this case, the image is not projected directly onto the windscreen, but onto a special pane located in front of the windscreen.

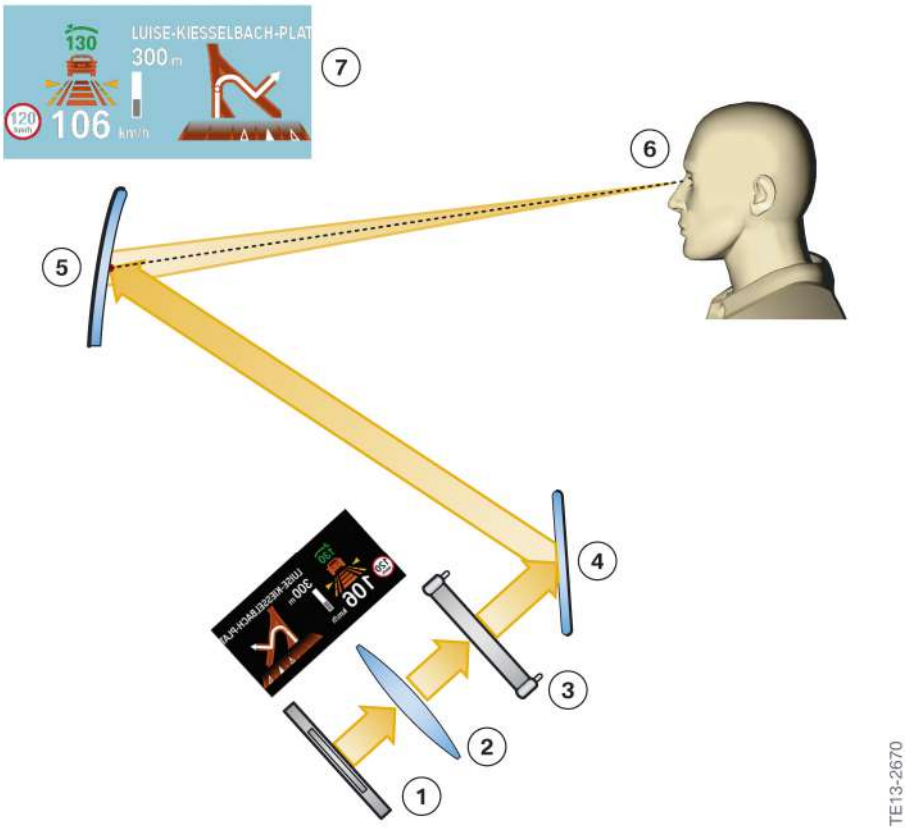
Advanced vehicle electrical sys.

8. Displays and Controls.



HUD with indicator pane

Function



Function HUD with indicator pane

Index	Explanation
1	Light source
2	Lens
3	TFT display
4	Plane mirror
5	Curved combination disc
6	Observer's point of vision
7	Projected image

Installation location

The installation location is again in the instrument cluster area.



F45 Installation location of HUD

8.1.3. Settings

Due to different seat positions, it may be necessary to adjust the image settings to match the individual driver. A major part of the settings can be completed using the controller.

More detailed settings need to be performed using the workshop information system.

The following settings can be performed using the controller on the Head-Up Display:

- Image content
- Brightness
- Height
- Rotation.

Additional geometric representations can be completed using the workshop information system.

Advanced vehicle electrical sys.

8. Displays and Controls.

8.2. Personal Profile

Personal profiles store the personal settings, such as the seat adjustment, radio stations, A/C settings, etc. in the respective control units. The settings are also assigned to the ID transmitter currently in the vehicle. Once the vehicle has been unlocked using the ID transmitter, the system identifies the settings that match the ID transmitter via a central control unit.



Personal Profile function

Profiles can be personalised. If the customer did **not** use their own ID transmitter to unlock the vehicle, they can still access their personal settings via the controller in the "Profiles" menu.

This system also offers benefits for the workshop. Many customers do not want their personal profiles to be changed. The service employee then has the option to select the "Guest" profile. Any changes they make to the settings then only affect the "Guest" profile. Before handing over the vehicle to the customer, the customer profile is re-activated; the customer's settings are automatically restored.

Personal profiles can be retrieved since the E87 via the ID transmitter. However, profile personalisation is only supported as of the F01/F02.



In the case of the F series, the workshop should activate the "Guest" profile to avoid changing the customer settings.

In E series vehicles with a personal profile, the workshop should unlock the vehicle with the mechanical key to avoid changing the customer settings.

8.2.1. Functional groups

The following systems can be personalised:

- Radio/Media
- Audio
- Navigation
- Favourite buttons
- Seat, mirror, steering column
- Camera settings (Surround View)
- Light, central locking system
- Head-Up Display
- A/C settings
- Language/units
- Assistance systems
- Driving experience switch
- Address book, etc.

8.2.2. Import and export

The customer can export the profiles from the vehicle and import them into a new vehicle. The export and import depends on the equipment level. For example, a profile from a vehicle with electrically adjustable seats, the seat adjustment cannot be imported into a vehicle with mechanically adjustable seats.

The export and import is either handled using a USB stick, or via an online ConnectedDrive connection.

8.3. Central information display

The central information display, CID, is the central display in vehicles. The CID is activate by the respective head unit via a special interface.

Advanced vehicle electrical sys.

8. Displays and Controls.



CID connection to the head unit

In current BMW vehicles, various sizes of CID are installed. The size depends on the model and equipment level.

In the following graphic, you can see the difference in the display sizes based on the example of a F12:



F12 CIDs compared

Index	Explanation
1	7" screen diagonal; resolution of 800 x 480 pixels
2	10.25" screen diagonal; resolution of 1280 x 480 pixels

The G11/G12 currently has the CID with the highest resolution of 1440 x 540 pixels.

8.3.1. Touchscreen

Since the G11/G12, the central information display is no longer just a display for graphically portraying content. The CID in the G11/G12 is a touch screen. This makes the CID a complete operating element which can be used to control all functions.



CID with touch screen

8.4. Controller

The controller is the central operating element for the vehicle functions that are selectable and configurable on the CID. 7 direct access keys with a static assignment are arranged around the controller. They directly open the respective menu.

In terms of control, the control panel supports various degrees of freedom for navigating in the menu structure:

- Turn
- Press
- Tilt.

8.4.1. Variants

Installation of the controllers depends on the equipment level. The following table shows the controllers installed for the various equipment levels:

Controller	Equipment
5-button	Radio Professional
7-button	Business navigation system
7 buttons with touch control panel	Professional navigation system

Controller BMW

The following table shows you the controllers currently deployed by BMW:

5-button	7-button	7 buttons with touch control panel	7 buttons with touch control panel (from G11/G12)
			

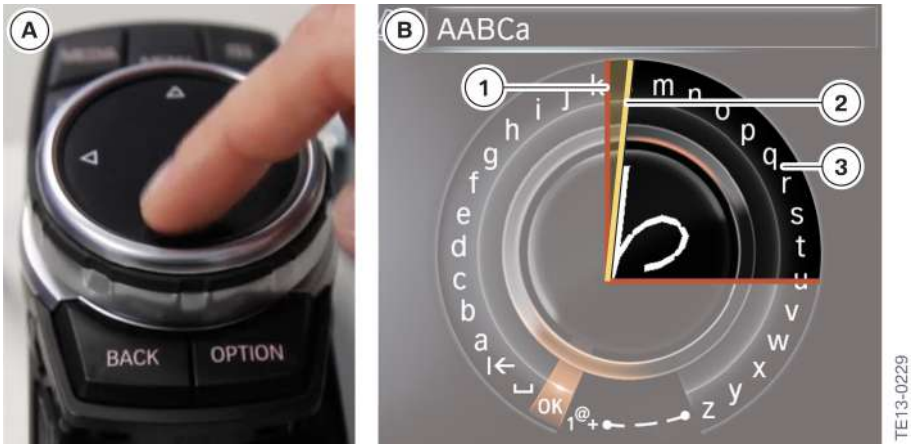
Touch control box

Since the F10 LCI, vehicles with the optional equipment "Navigation System Professional" have a controller with a touch control box installed.

In the touch control panel the customer can input location information for the navigation system or telephone numbers and contact details. In the map operation the map section for example can be moved and enlarged or reduced by finger movement. Using the touch control box the mouse pointer can also be moved for the "Internet" function.

Advanced vehicle electrical sys.

8. Displays and Controls.



Input controller with touch control panel

Index	Explanation
A	Input via controller with touch control box
B	Displays in central information display (CID)
1	Edge of angular range for input
2	"Angle" of input
3	Angular range for input

An additional control unit, the Touchbox TBX, is installed to evaluate the characters.



Signal path controller with touch control box

Index	Explanation
1	Central information display CID
2	Head Unit High
3	Controller with touch control panel
4	Touchbox (TBX)

In the G11/G12 and the successor vehicles, characters are no longer detected via a separate touchbox, but via a printed circuit board in the Head Unit High 2.

8.4.2. MINI

Controller

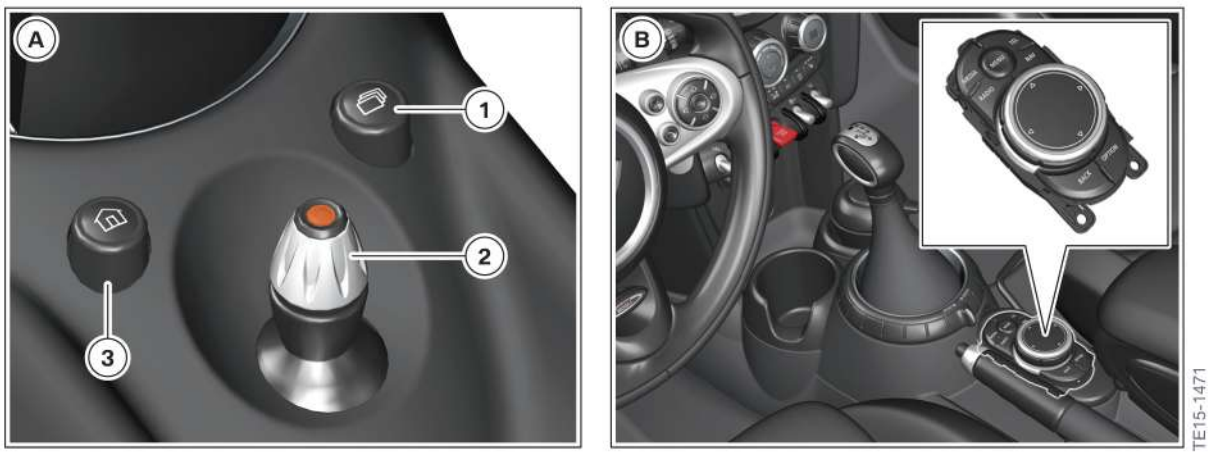
The following table shows you the controllers currently deployed by MINI:

5-button	7-button	7 buttons with touch control panel
		

MINI joystick

MINI vehicles up to the F56 have the MINI joystick installed instead of a controller.

The operation of the MINI joystick is similar to that of a controller (turn, push, tilt).



MINI joystick

Index	Explanation
A	MINI joystick (up to F56)
B	MINI Controller (from F56)
1	Mode button
2	MINI joystick
3	Menu button

Advanced vehicle electrical sys.

8. Displays and Controls.

8.5. Instrument panel

Different versions of the instrument cluster exist at BMW. However, the speed, engine speed and fuel filling level are always displayed in any instrument cluster. Every instrument cluster has a display on which additional content is displayed. This includes the on-board computer, the fuel consumption display, navigation or infotainment content.

The following table provides an overview of which instrument clusters are installed in the motor vehicles:

Vehicle	Instrument panel
1 Series/2 Series/X1	• 2 round instruments • 2.7" and 5.7” TFT display
3 Series/4 Series	• 4 round instruments • 2.7" and 5.7” TFT display
5 Series, 6 Series, 7 Series	• 4 round instruments • 5.7" and 9.2” TFT display • Multifunctional instrument display
X3/X4	• 4 round instruments • 2.7", 5.7" and 9.2” TFT display
X5/X6	• 4 round instruments • 2.7", 5.7" and 9.2” TFT display • Multifunctional instrument display
Z4	• 2 round instruments • Liquid crystal display

8.5.1. 2 round instruments

The 2 round instruments contain the speed and engine speed displays. The fuel gauge is located below the speed indicator. The instrument clusters have an integrated TFT display. The size of the TFT display depends on the vehicle equipment.

Advanced vehicle electrical sys.

8. Displays and Controls.



instrument cluster BMW 2 round instruments

Index	Explanation
1	Instrument cluster with 2.7" TFT display
2	Instrument cluster with 5.7" TFT display

Instrument clusters in E series have 2 round instruments and a liquid crystal display located at the centre.



Instrument cluster BMW E series liquid crystal display

8.5.2. 4 round instruments

Instrument clusters with 4 round instruments have the following integrated displays:

- Speedometer
- Rev counter
- Fuel gauge
- Engine temperature indicator



Instrument cluster BMW 4 round instruments

Index	Explanation
1	Instrument cluster with 2.7" TFT display
2	Instrument cluster with 5.7" TFT display
3	Instrument cluster with 9.2" TFT display

The size of the TFT display depends on the optional equipment installed in the vehicle.

8.5.3. Multifunctional

In the case of multifunctional instrument displays, the major part of the instrument cluster consists of a TFT display. The high version of the G11/G12 has an instrument display where all content is exclusively shown on a TFT display.



G11G12 Multifunctional instrument display

8.5.4. MINI

The instrument cluster in current MINI models(as of F56) is driver oriented and installed on the steering column.

Advanced vehicle electrical sys.

8. Displays and Controls.



MINI instrument cluster

Index	Explanation
1	Rev counter
2	Speedometer
3	Warning and indicator lights
4	Fuel gauge
5	On-board computer in the TFT display (2.7")

The instrument cluster in older MINI models (before F56) is installed at the centre above the centre stack in the instrument panel.

Advanced vehicle electrical sys.

8. Displays and Controls.



MINI Countryman instrument cluster

Index	Explanation
1	Engine speed display with indicator lights
2	Speed indicator and fuel gauge

Advanced vehicle electrical sys.

9. Infotainment.

9.1. Media playback

9.1.1. CD/DVD

CD/DVD drive in radio/head unit

Most BMW vehicles have a CD or DVD drive built into the radio or head unit. However, the CD drive is optional equipment with the following models:

- MINI (from F56)
- BMW (from F45/F46 and F48)

The CD/DVD drives can play back MP3 files. It is possible to play back movies from a DVD using the DVD drive.



CD player

Index	Explanation
1	CD-/DVD drive

DVD changer

A 6x DVD changer is installed in the luggage compartment or the glove box as optional equipment. Optional equipment "DVD changer" can longer be ordered since F45. However, these vehicles already have a USB audio interface as standard equipment.

In the following graphic, you can see the installation location of the DVD changer, based on the example of a F20:



F20 DVD changer

The shape of the CDs/DVDs must meet the specifications of the geometrical features (diameter 120 mm). If CDs/DVDs with a different geometry are used (not round in shape, different size), they can damage the drive.

9.1.2. USB audio interface

AUX-IN port

Audio files are transmitted to the head unit via the AUX-IN port. The media source cannot be controlled via the head unit here. All settings must be made on the media source (MP3 player, iPod, etc.). The exception is volume control.

USB interface

In current BMW vehicles (since F45), a USB audio interface is already included as standard equipment.

USB sticks, MP3 players, or smartphones can be connected to the USB audio interface.

In addition to music or video files, the USB audio interface is used to update Bluetooth compatibility or import and export profiles.



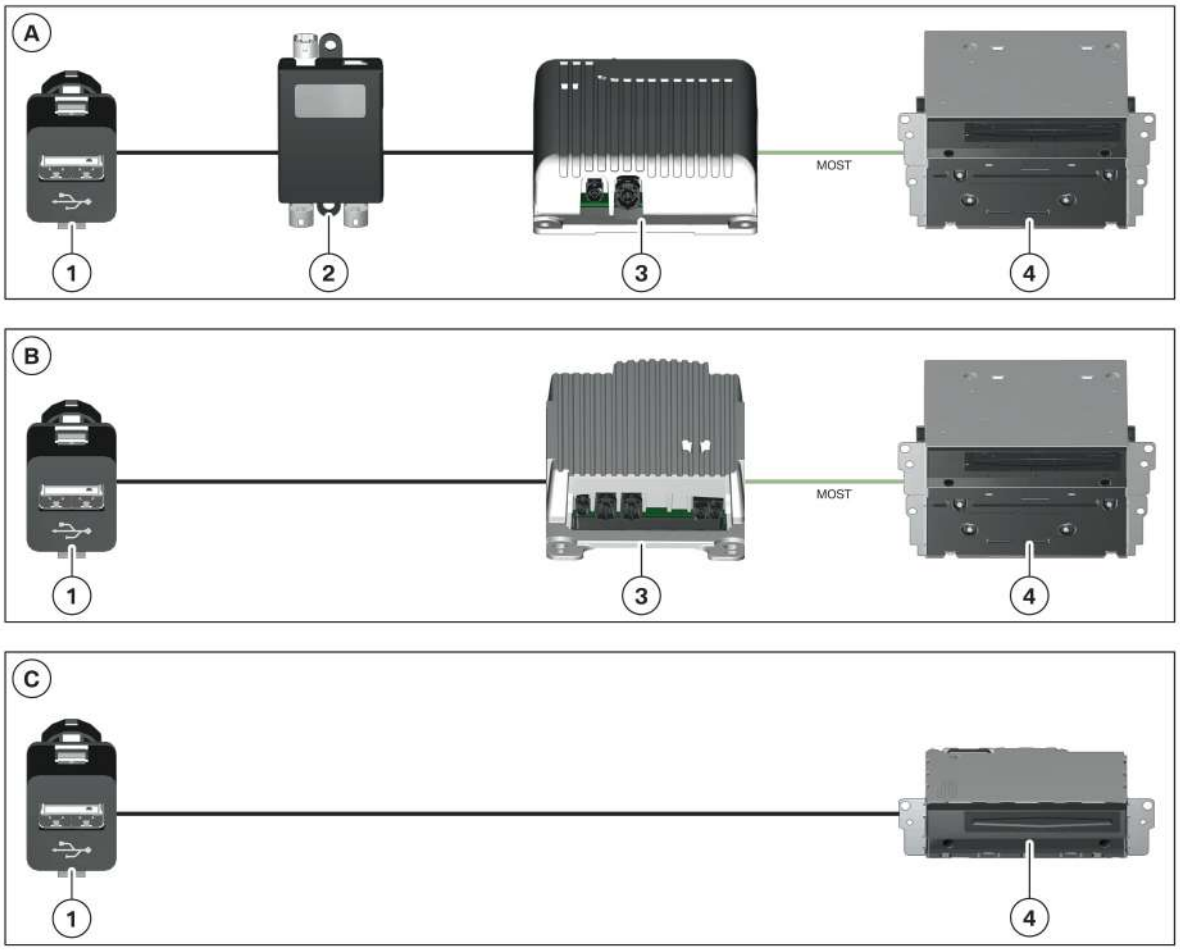
USB audio interface

Index	Explanation
1	AUX-IN port
2	USB audio interface

⚠

Only approved USB devices can be connected. USB hard disks or USB devices with too high a power consumption must not be connected.

The USB audio interface is either directly connected to the head unit, or connected via a special control unit, depending on the vehicle.The first vehicle at BMW to use a direct connection of the USB audio interface to the head unit is the F25.



USB audio interface connection

Index	Explanation
A	Vehicles up to 2010 (via USB hub and interface box High ULF-SBX-H
B	Vehicles as of 2010 (via Combox)
C	Vehicles as of the F25 (first direct audio playback through the head unit at BMW; for MINI as of Head Unit Basic 2 (CHAMP 2)).
1	USB audio interface
2	USB hub
3	Up to 2010 interface box High, SBX high; as of 2010 Combox
4	Head unit

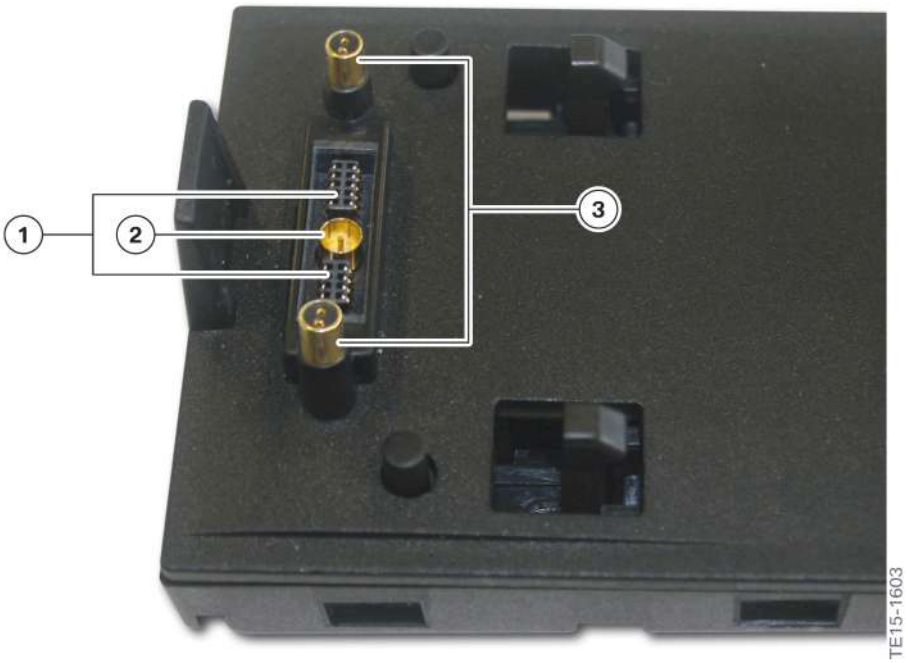
As of 2010, the interface box High was replaced by a new control unit, the Combox. The use of the Combox allowed the USB hub to be dropped.

If a USB device does not work in the vehicle, connect the USB device to a computer to test the function there.

In case of a fault, the USB audio interface can be checked with the IMIB R2 measuring equipment.

Snap-in adapter

Audio playback can also occur via the MP3 player integrated in a mobile phone.For audio playback, the mobile phone would need to be taken out of the snap-in adapter and connected to the USB audio interface using a USB cable. To save the customer this work, a snap-in adapter with integrate USB audio interface was introduced as optional equipment on F01/F02. This special snap-in adapter is identifiable by its golden pins.



Snap-in adapter-extended smartphone connection

Index	Explanation
1	18-pin multi-pin connector on the base plate (AUX-In, voltage supply)
2	Aerial port
3	USB port

9.1.3. Music collection

Some of the memory capacity on the hard disk of the head unit High (since F01/F02 with navigation system Professional) is reserved for audio files. The customer can store and manage music on the existing hard disk partition. The memory capacity for the music collection depends on the respective head unit (from 8 GB to currently 26 GB). The audio files can be stored on the hard disk using 3 approaches:

- Copying an audio CD via the drive in the head unit
- Copying MP3 files from CD/DVD via the drive in the head unit (ripping; conversion to a special BMW AAC format)
- MP3-Copying files from the USB stick via the USB interface.

Vehicles in which the head unit CIC is installed, need to transfer the MP3 files from the USB stick via the USB interface in the glove box. The USB audio interface is used for this purpose for newer head units.



The customer has the option of backing up their music collection on a USB stick. They should repeat the backup at specific intervals. If the head unit is faulty, there is no way to restore the music collection.



Due to copyright law the Service employee is not allowed to perform the backup for the customer. However, the Service employee is allowed to instruct the customer when he is performing the backup procedure.

9.1.4. Bluetooth audio streaming

Since March 2010 it is possible to use a Bluetooth connection to play back audio files from a smartphone or an MP3 player. The head unit is used to control this.



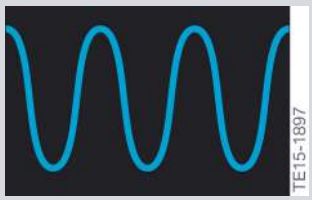
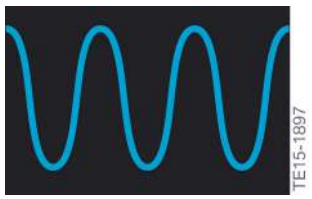
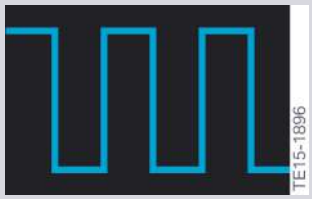
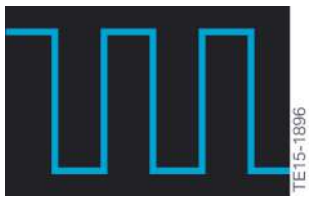
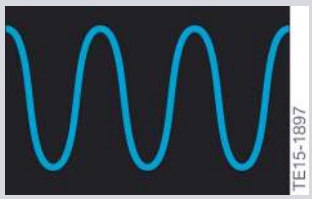
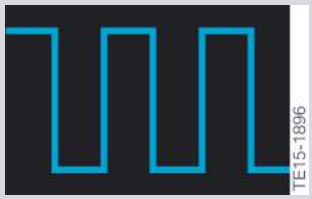
Bluetooth audio streaming

In case of a fault, the Bluetooth connection can be checked with the IMIB R2 measuring equipment.

To ensure optimum functionality of audio player via Bluetooth, only devices recommended by BMW may be used. A list is available on http://www.bmw.com/bluetooth.

9.1.5. Radio

The following table provides an overview of the radio types that can be received at BMW. Reception of the various radio type depends on the country and equipment.

Radio	Reception type
AM Amplitude modulation	Analogue 
FM Frequency modulation	Analogue 
DAB Digital Audio Broadcasting	Digital 
SDARS Satellite Digital Audio Radio Service	Digital 
IBOC In Band On Channel	Analogue  and digital 

AM and FM signals are transmitted and received using analogue methods.

DAB (Digital Audio Broadcasting) is digital radio.

IBOC (In Band On Channel) is hybrid radio. The signals are sent both digitally and using analogue methods. This radio is also referred to as HD radio.

SDARS is a radio that receives satellite signals.

9.2. Telephone

9.2.1. Bluetooth

Since 2002, the hands-free system in the vehicle has relied exclusively on a Bluetooth connection.

Snap-in adapter

For improved reception and to charge the mobile phone battery, a snap-in adapter is available as an accessory for most mobile phones. The snap-in adapter offers a connection to the vehicle's external aerial via the base plate. The battery in the mobile is charged via a power supply.



Snap-in adapter

Control unit

Since the introduction of Bluetooth telephony at BMW (2002), a control unit has been required for this function. A Bluetooth aerial, which the mobile phone uses to communicate, is connected to the control unit. The following list provides an overview of the control units used previously.

- Universal charger and hands-free unit ULF
- Telematic Control Unit TCU
- Combox.

Advanced vehicle electrical sys.

9. Infotainment.



F01 Bluetooth telephony function

Index	Explanation
1	Mobile phone in the snap-in adapter
2	Telematic Control Unit TCU
3	Roof aerial

Since the introduction of the head unit High (2012), telephony has been integrated into the head unit. A separate control unit is not required for the hands-free system.

9.3. Navigation system

Navigation systems not only guide the driver to the destination more quickly and comfortably, but also provide support with extensive and easy-to-use functions in every driving situation.

9.3.1. Function

The following section explains the function of a navigation system in a greatly simplified way.

The navigation system receives a signal transmitted by GPS satellites. Based on these signals, the navigation system computes the current vehicle position. A route is computed by referring to the map data and continually compared with the current position using GPS.



Schematic function of navigation

At BMW, there are currently two different navigation systems. The Business navigation system and the Professional navigation system.

The control unit responsible for the navigation function resides in the head unit.

9.3.2. Business navigation system

The Business navigation system is characterised by the following features:

- Fixed 6.5" colour display
- 2D map view
- Controller
- Hands-free system
- USB audio interface
- Radio Professional
- CD player
- Favourite Buttons.



Business navigation system

9.3.3. Professional navigation system

The Professional navigation system is characterised by the following features:

- Fixed 8.8" or 10.25" colour display
- Split-screen function (main and assistance window)
- 3D map view
- Controller (with touch control box depending on the vehicle)
- Hands-free system
- USB audio interface
- Radio Professional
- DVD drive
- 20 GB storage capacity for audio files
- Favourite Buttons.



Professional navigation system

9.3.4. Map data

To be able to function, each navigation system in the vehicle needs map data. Based on the map data, the navigation system computes a route. Because today's map data do not only include roads, but other information such as speed restrictions, altitudes, bend radii, etc. the map data volume has now reached the gigabyte range. For this reason, the map data cannot cover the complete world map. The map data are broken down into regions depending on the size of the country and the infrastructure (e.g. map of Europe or map of the USA).

Customers or BMW Service can update obsolete map data. Depending on the vehicle and navigation system, the map data is updated via a CD, DVD or USB stick. For current models (since F23), some of the map data is updated automatically online up to four times a year. A complete update of the map data is performed either by BMW Service or the customer him-/herself. The customer purchases the map data from the BMW service workshop depending on the navigation system and vehicle. The map data can now also be downloaded via the ConnectedDrive portal.

9.4. Rear seat entertainment system

Entertainment sources (radio, TV, DVD) are played back in the rear passenger compartment via the optional equipment rear seat entertainment. Audio playback is either via the speakers or via headphones. Two different media sources can be used by the two rear compartment displays.



G11/G12 Rear seat entertainment

9.4.1. Rear compartment controller

In the F01/F02, rear seat entertainment is controlled via a rear-compartment controller. For other vehicles with rear seat entertainment (e.g. F10, F15 etc.) control is via a remote control.



Rear-compartment controller F01/F02

9.4.2. BMW Touch Command

The rear seat entertainment in the G11/G12 is controlled via a 7" tablet, the BMW Touch Command. The BMW Touch Command is connected to the head unit via the a special WLAN connection.



G11/G12 BMW Touch Command

9.5. Aerial systems

The following table features an overview of the aerial systems used by BMW:The aerial installation locations differ from vehicle to vehicle. The following table shows one possible installation location.

Advanced vehicle electrical sys.

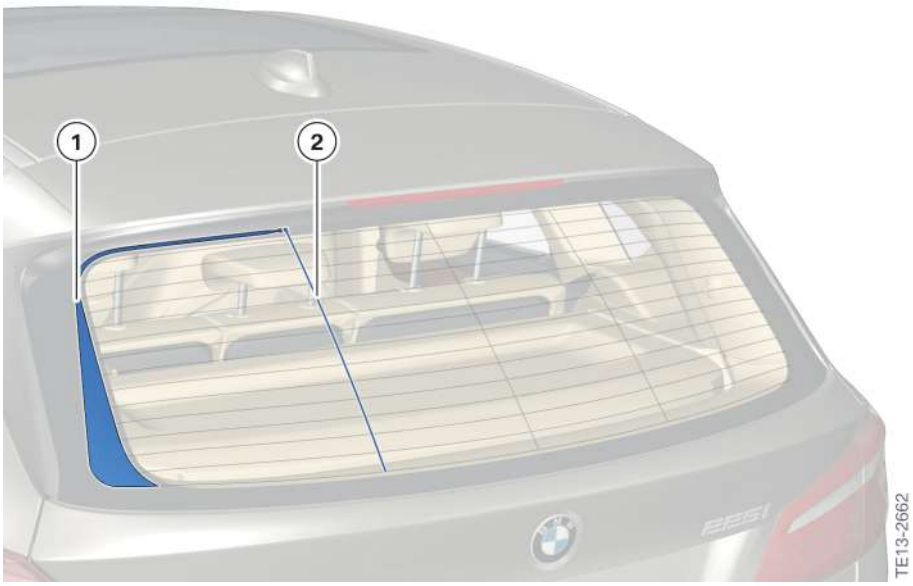
9. Infotainment.

Aerial	System	Possible installation location
FM aerial	Radio	• Rod aerial • Lower
AM aerial	Radio	• Rod aerial • Lower
DAB aerial	Radio	DAB L band in the roof-mounted aerial DAB band III in the rear window
SDARS aerial	Radio	Roof aerial
GPS aerial	Navigation system	Roof aerial
Remote control services aerial	Body Domain Controller	Luggage compartment, right
Telephone aerial	Telephone	Roof aerial
Bluetooth aerial	Telephone	Near the head unit
Emergency call aerial (back-up aerial)	Telematics services	Below the rear seat bench
VICS aerial	Japan navigation system	Inside mirror



F45 components of the aerial system

Index	Explanation
1	Roof aerial
2	Wave trap
3	Interference suppression filter
4	AM aerial
5	Aerial amplifier
6	Emergency call aerial



F45 aerial system, rear window

Index	Explanation
1	Rear window aerial, FM2
2	Rear window aerial, FM1/DAB Band III

9.6. Speaker systems

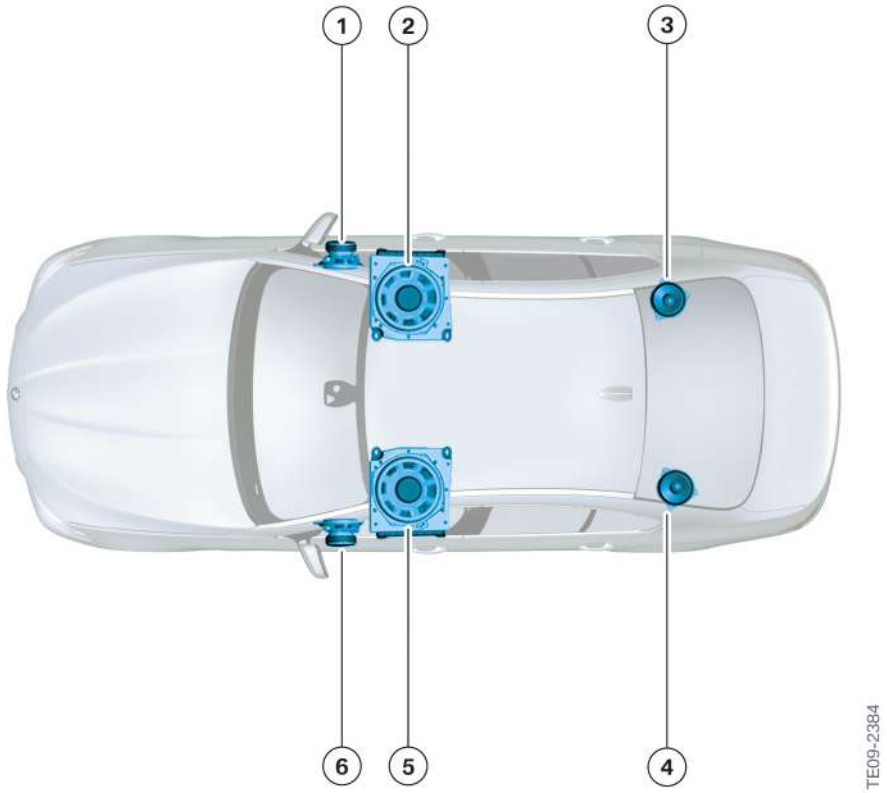
Various speaker systems are available from BMW.

- Stereo system
- Hi-Fi system
- Top HiFi system

The stereo system is included as standard equipment in most vehicles.The HiFi and Top HiFi systems are optional equipment.

9.6.1. Stereo system

The following graphic shows the speakers of a stereo system in a F10. All speakers are driven by the amplifier built into the head unit.



F10 Stereo system

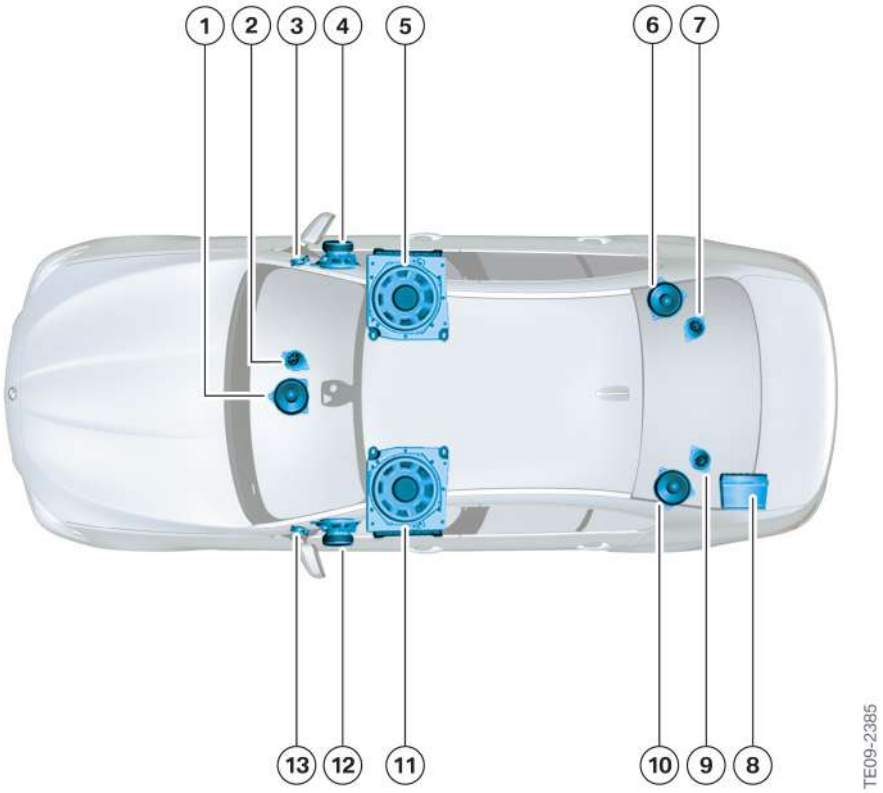
Index	Explanation
1	Mid-range speaker, front right
2	Bass speaker, right
3	Mid-range speaker, rear right
4	Mid-range speaker, rear left
5	Bass speaker, left
6	Mid-range speaker, front left

9.6.2. HiFi

The speakers and the amplifier of the Hi-Fi system in a F10 are illustrated in the following graphic. The amplifier does not have diagnostic capability.

Advanced vehicle electrical sys.

9. Infotainment.

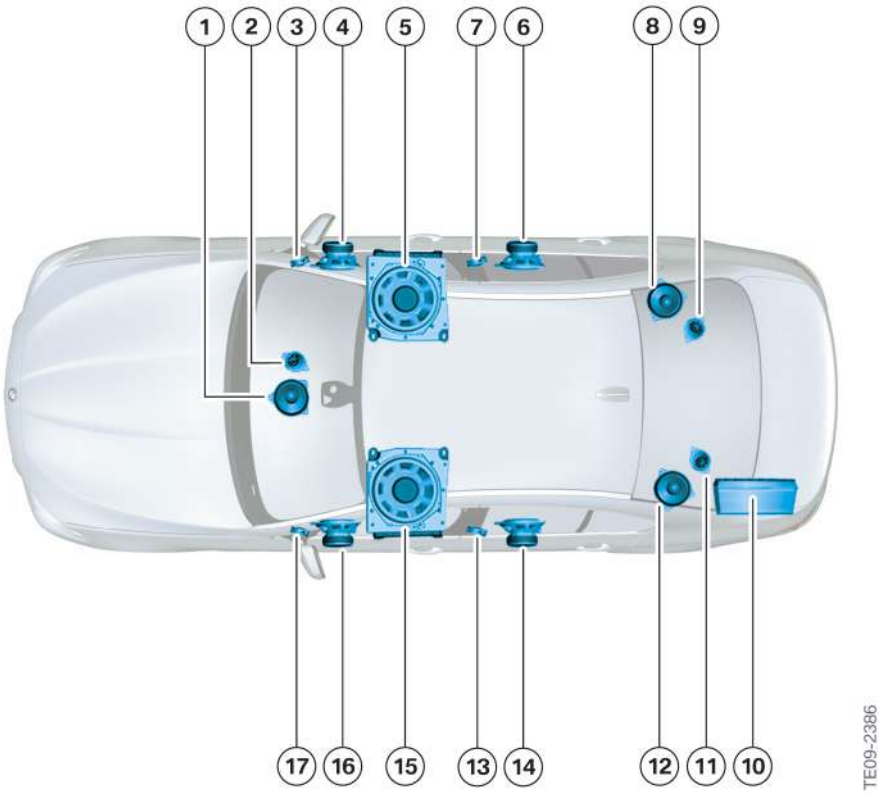


F10 Hi-Fi system

Index	Explanation
1	Tweeter, front centre
2	Mid-range speaker, front centre
3	Tweeter, front right
4	Mid-range speaker, front right
5	Bass speaker, right
6	Mid-range speaker, rear right
7	Tweeter, rear right
8	Hi-Fi amplifier
9	Tweeter, rear left
10	Mid-range speaker, rear left
11	Bass speaker, left
12	Mid-range speaker, front left
13	Tweeter, front left

9.6.3. Top HiFi

The speakers and the amplifier of the Professional (Top HiFi) high fidelity system in a F10 are illustrated in the following graphic. The amplifier is connected to a bus system and the system thus has diagnostic capability.



F10 Top HiFi system

Index	Explanation
1	Tweeter, front centre
2	Mid-range speaker, front centre
3	Tweeter, front right door
4	Mid-range speaker, front right door
5	Bass speaker, right
6	Mid-range speaker, rear right door
7	Tweeter, rear right door
8	Mid-range speaker, rear right
9	Tweeter, rear right
10	Top HiFi amplifier
11	Tweeter, rear left
12	Mid-range speaker, rear left

Index	Explanation
13	Tweeter, rear left door
14	Mid-range speaker, rear left door
15	Bass speaker, left
16	Mid-range speaker, front left
17	Tweeter, front left

Advanced vehicle electrical sys.

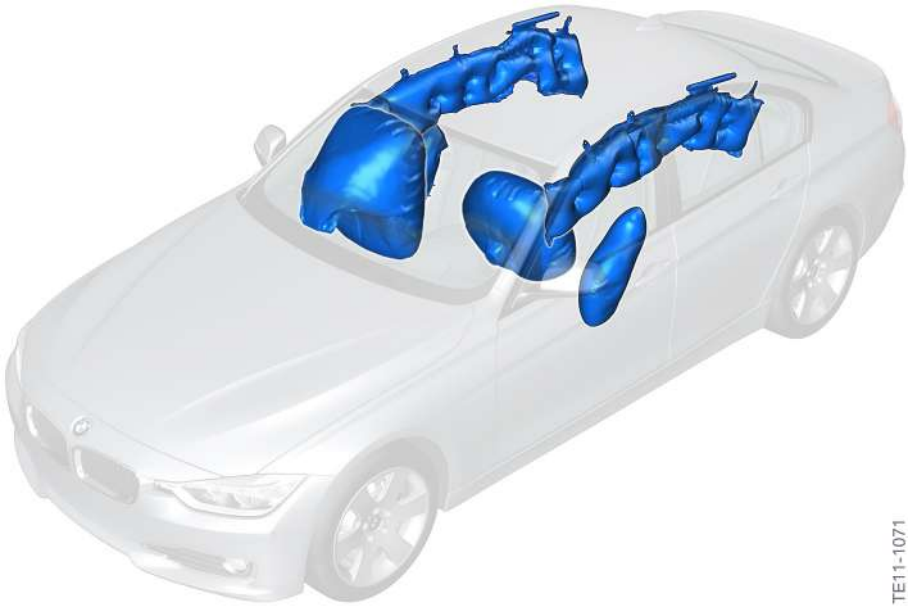
10. Passive Safety.

"Passive safety" means all design measures that serve to protect the occupants from injury in case of an accident, or to mitigate the consequences of an accident. But the protection of other road users in the event of an accident also falls under the term "passive safety".

The deformation zones at the front, rear and side of the body, as well as the deformation-resistant passenger cell, are also part of passive safety. In addition, airbag and restraint systems are also an important element of the passive safety of a vehicle.

10.1. Airbags

The airbag and restraint systems are aligned precisely to the deformation behaviour of the body. Depending on the type and severity of the accident, they are responsible for deploying the correct airbag with the required restraining effect at the exact time. Airbags and belt tensioners that are not required are not activated by the intelligent system. They continue to be available in the event of a possible second collision.



F30 Airbags

Advanced vehicle electrical sys.

10. Passive Safety.

10.2. Control units, sensors and switches



F25 sensors, switches and control units of the airbag and restraint systems

Index	Explanation
1	Airbag front sensor, right and left (US version)
2	Switch for front passenger airbag deactivation
3	Lateral and longitudinal acceleration sensor, B-pillar, right and left
4	Seat occupancy mat (US version: CIS mat)
5	Integrated Chassis Management (ICM) (integrated sensors for roll-over detection, etc.)
6	Seat belt buckle switch for driver's and passenger belt
7	Airbag sensor, front door, driver's and passenger's side (US version)
8	Crash Safety Module (ACSM)

Advanced vehicle electrical sys.

10. Passive Safety.

The number and version of the control units, sensors and switches of the airbag and restraint system vary depending on the national-market version. Information on the control units, sensors and switches must therefore be taken from the vehicle-specific repair instructions.

10.2.1. Control units

There are often additional sensors located in the control units. For example, the sensors for the roll-over detection of the F25 are located in the Integrated Chassis Management (ICM). If control units with integrated sensors are moved adversely when a battery is connected, this may cause the activation of the individual airbags and belt tensioners. This is one of many reasons why it is imperative to disconnect the battery before the removal of control units.

10.2.2. Airbag sensor, door

Airbag sensors are also installed in the driver's and front passenger door depending on the national-market version. The airbag sensors in the doors provide support in the event of a side impact detection, by measuring the internal door pressure. In addition to the lateral acceleration values, the pressure in the door cavity also increases in the event of a side collision.



In order to ensure the correct function of the airbag sensors in the event of a side collision, it is necessary to pay attention to the correct installation of all sealing measures and the door trim panels during the repair.

10.2.3. Acceleration sensors

The lateral and longitudinal acceleration sensors provide support in the event of the detection of a front-end, side and rear-end collision. In general, at least two crash signals are required from different sensors to trigger the activation of airbags and belt tensioners.



All testing and installation work at airbag and restraint systems can only be performed by experienced and trained personnel.

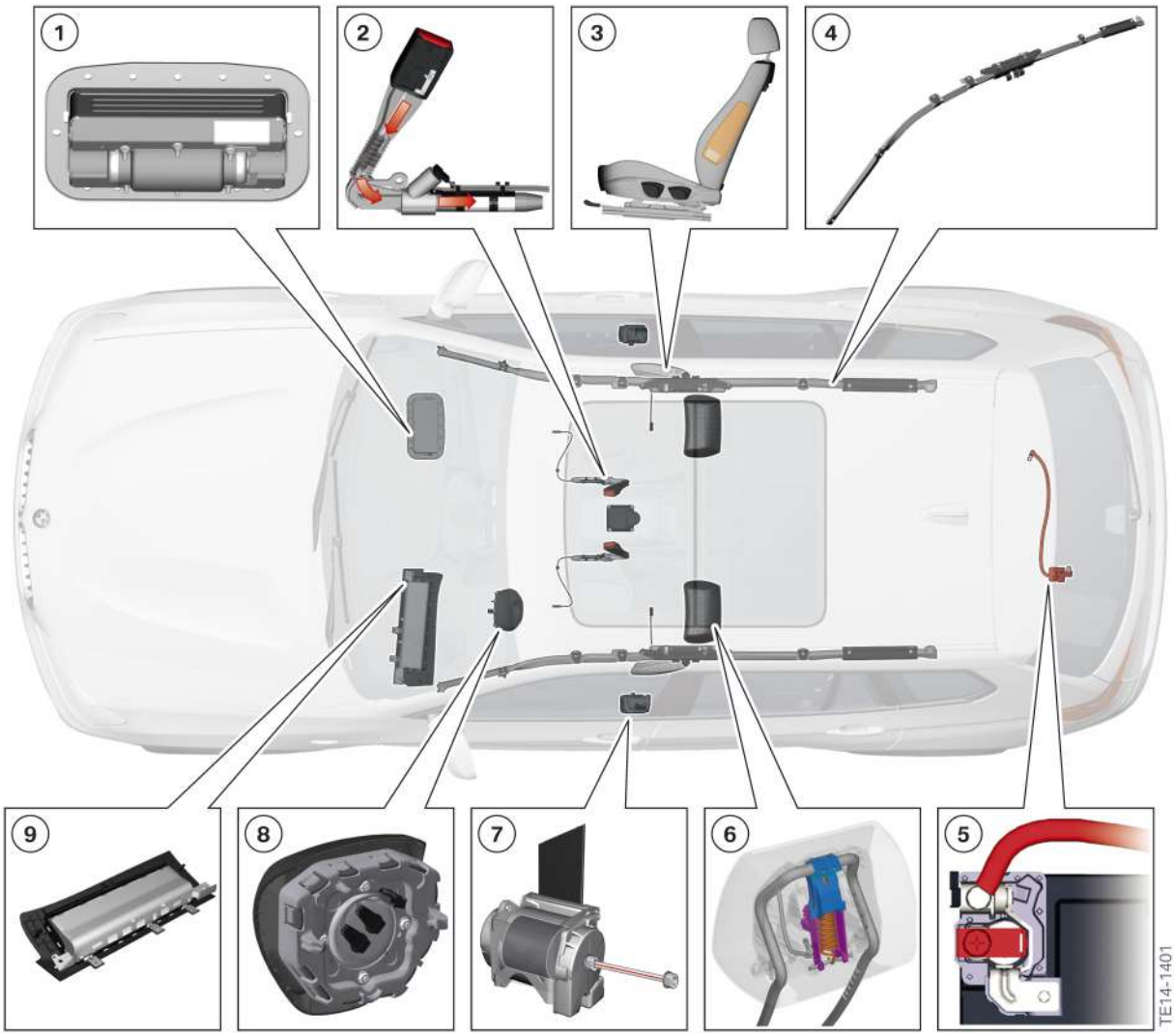
For all work on components of airbag and restraint systems, the information in the workshop information system, which is updated on a daily basis, must always be observed.

The battery must always be disconnected before the mounting of control units, sensors and gas generators is removed, or control units, sensors and gas generators are connected or disconnected. Otherwise, it may cause fault messages, and even undesired activation of airbags or belt tensioners.

Advanced vehicle electrical sys.

10. Passive Safety.

10.3. Pyrotechnic actuators



F25 pyrotechnic actuators of airbag and restraint systems

Index	Explanation
1	Front passenger airbag
2	Belt tensioner for driver's and passenger belt
3	Side airbag for driver's and front passenger seat
4	Head airbag for driver's and passenger's side
5	Safety battery terminal SBK
6	Crash-active headrest, driver's and front passenger seat
7	Retractor tensioner for driver's and passenger belt
8	Driver's airbag
9	Knee airbag for driver's and passenger's side (US version)

10. Passive Safety.

Airbags and belt tensioners are pyrotechnic objects. Their handling and transport are controlled differently depending on the national-market version.



All testing and installation work at airbag and restraint systems can only be performed by experienced and trained personnel.

For all work on components of airbag and restraint systems, the information in the workshop information system must always be observed.

Before all installation and removal work at airbag and restraint systems, the battery must be disconnected and the negative terminal, as well as the terminal, covered.

10.4. Pedestrian impact severity mitigation

Various measures are applied in order to reduce the severity of the injuries in the event of accidents with pedestrians.

10.4.1. Passive pedestrian protection

Passive measures for the protection of pedestrians in the event of accidents include:

- Engine compartment lid with flexible design
- Engine compartment lid hinge with mechanical release mechanism
- Module carrier of side panel modules with flexible design.

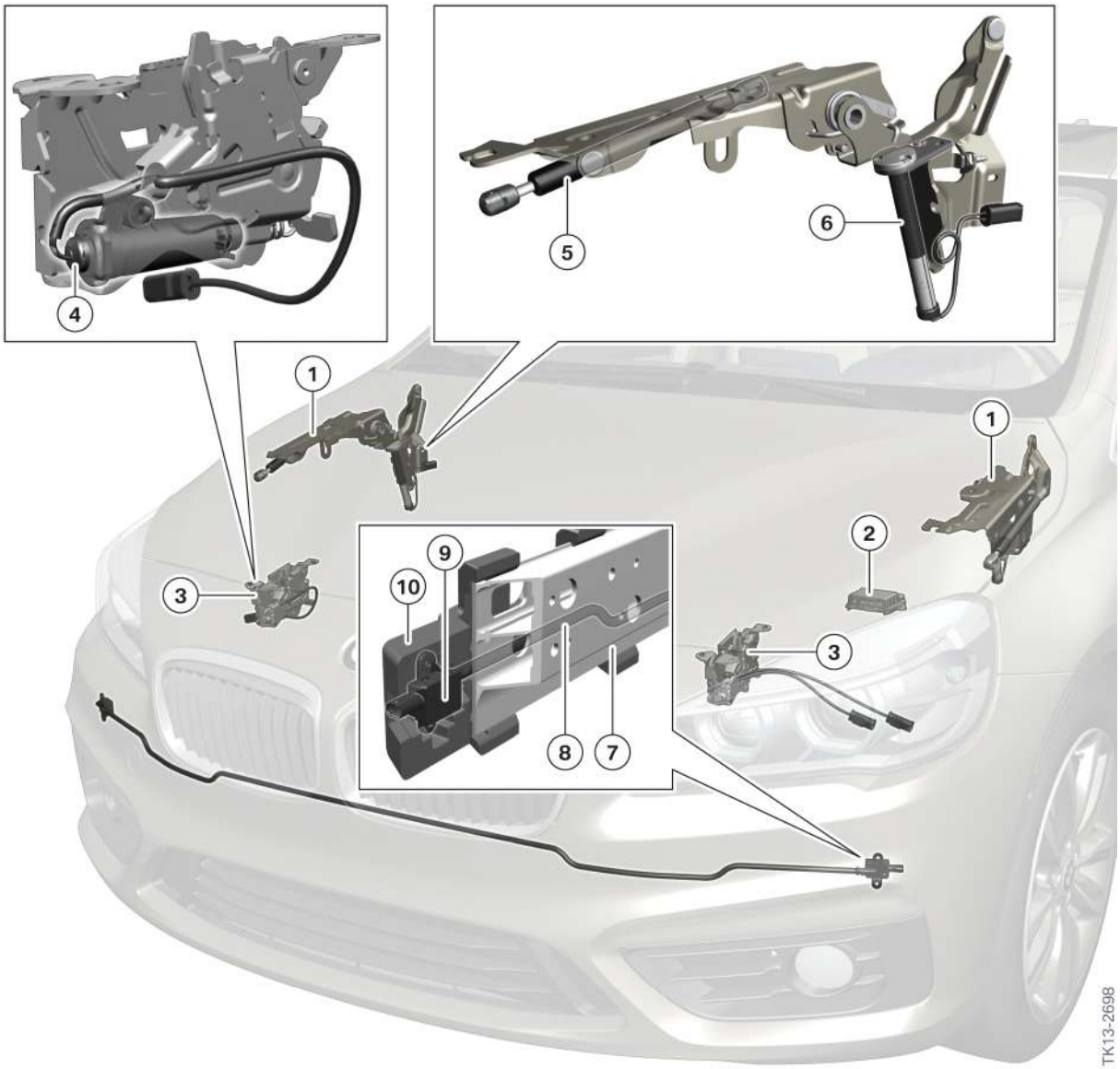
10.4.2. Active engine compartment lid

In the event of a collision with a pedestrian, for example, the engine compartment lid is raised. An enlarged crumple zone arises between the engine compartment lid and engine, which reduces the effects of the accident.

The raising of the engine compartment lid is activated by pyrotechnic actuators at the engine compartment lid hinges and catches.

Advanced vehicle electrical sys.

10. Passive Safety.



F45 active engine compartment lid

Index	Explanation
1	Engine compartment lid hinge (with actuator)
2	Crash Safety Module (ACSM)
3	Engine compartment lid catch (with actuator)
4	Actuator
5	Gas pressure springs
6	Actuator

Index	Explanation
7	Bumper support
8	Pressure hose
9	Pressure sensor
10	Impact absorber

10.4.3. Sensors

Sensors

A collision is detected from a certain driving speed by separate sensors. These are mounted behind the bumper. Different sensor technologies are used depending on the vehicle:

- Optical fibre
- Acceleration sensors
- Optical fibre and additional acceleration sensors
- Pressure sensors.

The sensors are always located in the immediate area of the front bumper. The information in the workshop information system must always be observed when handling the various sensors.



All testing and installation work at pyrotechnic components of the active engine compartment lid can only be performed by experienced and trained personnel.

The battery must be disconnected before all installation and removal work at the engine compartment lid hinges or catches.

Before carrying out adjusting procedures at the engine compartment lid, the battery must be disconnected by disconnecting the negative terminal from the vehicle electrical system.

Information on the various systems for pedestrian protection can be found in the workshop information system.

The information in the workshop information system must be observed.

10.5. Functions of Crash Safety Module

The function of the Crash Safety Module is to permanently evaluate all sensor signals in order to identify a crash situation. Based on the sensor signals and their evaluation, the Crash Safety Module (ACSM) identifies the direction of the crash and the severity of the impact.

The Crash Safety Module (ACSM) evaluates the sensor information and then initiates the relevant measures for selective activation of the necessary restraint systems.

At the same time the Crash Safety Module (ACSM) monitors the system itself and indicates when it is ready for operation by switching off the airbag indicator light.

If an error occurs during operation this is stored in a fault memory The fault memory can be read out for diagnosis purposes.

If a crash situation is detected, a crash message is sent to the other bus users in the data bus network as notification. The relevant control units respond to this signal by executing their own activities according to the severity of the crash.

The activities include:

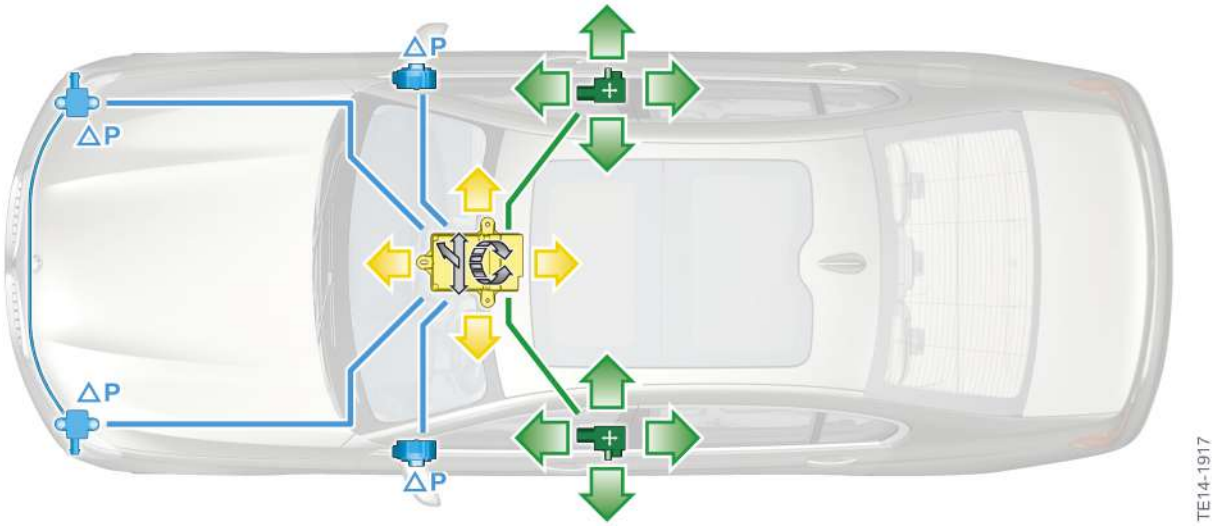
- Opening the central locking system
- Activating the hazard warning flasher
- Switching on the interior light
- Making an emergency call.

An additional function of the Crash Safety Module is the acoustic seat belt warning that reminds the driver using visual and acoustic signals to fasten his seat belt. With help of a seat belt buckle switch it is determined whether the driver has fastened his seat belt. Furthermore, for vehicles in the US version the position of the driver and the front passenger seat is monitored.

10.6. Crash detection for European versions of vehicles

The European version of the vehicles is equipped with the following sensors:

- one lateral and one longitudinal acceleration sensor in the B-pillars (green)
- one airbag sensor to monitor the pressure in each of the front doors (blue)
- one lateral and one longitudinal acceleration sensor in the ACSM (yellow)
- one roll rate sensor in the ACSM (yellow)
- one vertical acceleration sensor in the ACSM (yellow)
- one pressure hose with two pressure sensors behind the front bumper (blue).



G11G12 Impact detection

The longitudinal acceleration sensors can measure both positive and negative acceleration. A front-end or rear-end crash can be detected with the assistance of these signals.

A side-on crash can be detected with the assistance of the lateral acceleration sensors.

Rollover is detected by the roll rate sensor and vertical acceleration sensor in the Crash Safety Module (ACSM) .

A crash whose effective direction does not correspond to the longitudinal or transverse axis of the vehicle is detected by a combined lateral and longitudinal acceleration.

10.7. Rollover protection system

Open vehicles, convertibles and roadsters, have a long tradition at BMW. The statistics at BMW Accident Research reveal that a rollover occurs in the sequence of events in approx. 20% of serious accidents. Convertibles and roadsters are no exception. Due to the lack of a roof, these rollovers can have serious consequences for the occupants. For this reason, sufficient survival space must be created for the occupants of open vehicles. In the case of BMW vehicles, the survival space is mainly created by means of a reinforced windscreen frame, the rear end shape and the rollover protection bar. Fixed rollover protection is used for purist roadsters. This means that two rollover protection bars are permanent visible. To avoid interrupting the elegant line of the convertibles, extending rollover protection bars are used. This means, the bars are hidden and only appear in case of a vehicle rollover.

Advanced vehicle electrical sys.

10. Passive Safety.

10.7.1. Fixed rollover protection bar



Fixed rollover protection bar E89

10.7.2. Extending rollover protection system



F23 Fully recessed rollover protection system

Index	Explanation
1	Serrated strip
2	Interlock pawl
3	Electromagnetic actuator



Bayerische Motorenwerke Aktiengesellschaft
Händlerqualifizierung und Training
Röntgenstraße 7
85716 Unterschleißheim, Germany